



Analysing Factors Associated Early Readmission of Diabetic Patients

GROUP 2

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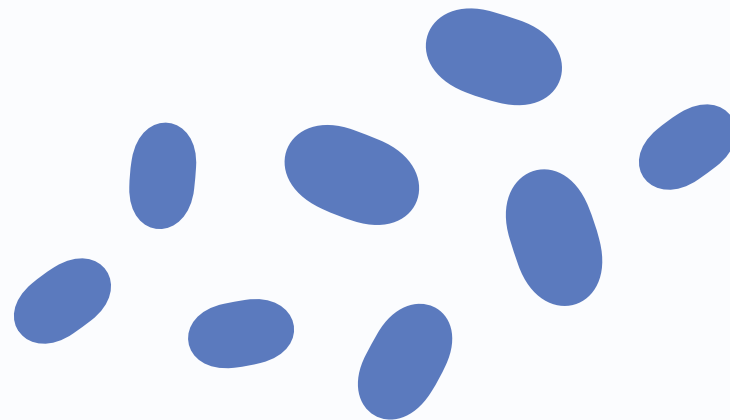
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Problem Statement

Identify statistically significant clinical and demographic factors associated with early hospital readmission (≤ 30 days) among diabetic patients, and develop an interpretable regression-based model to estimate early readmission risk.





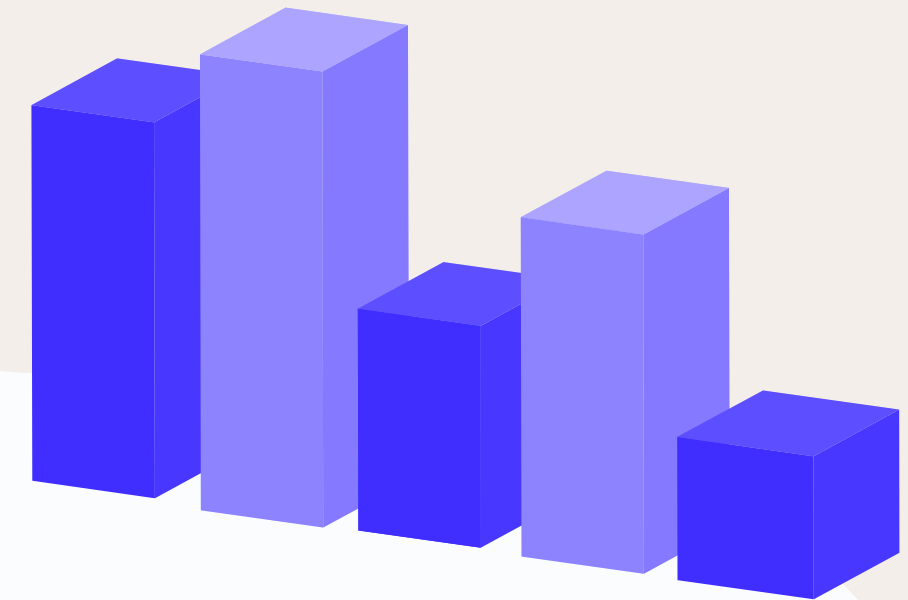
Dataset Details

Name: 130-US Hospital dataset (1999 - 2008)

Source: Kaggle

No. Of Features: 50

No. Of Observations: 1,00,000

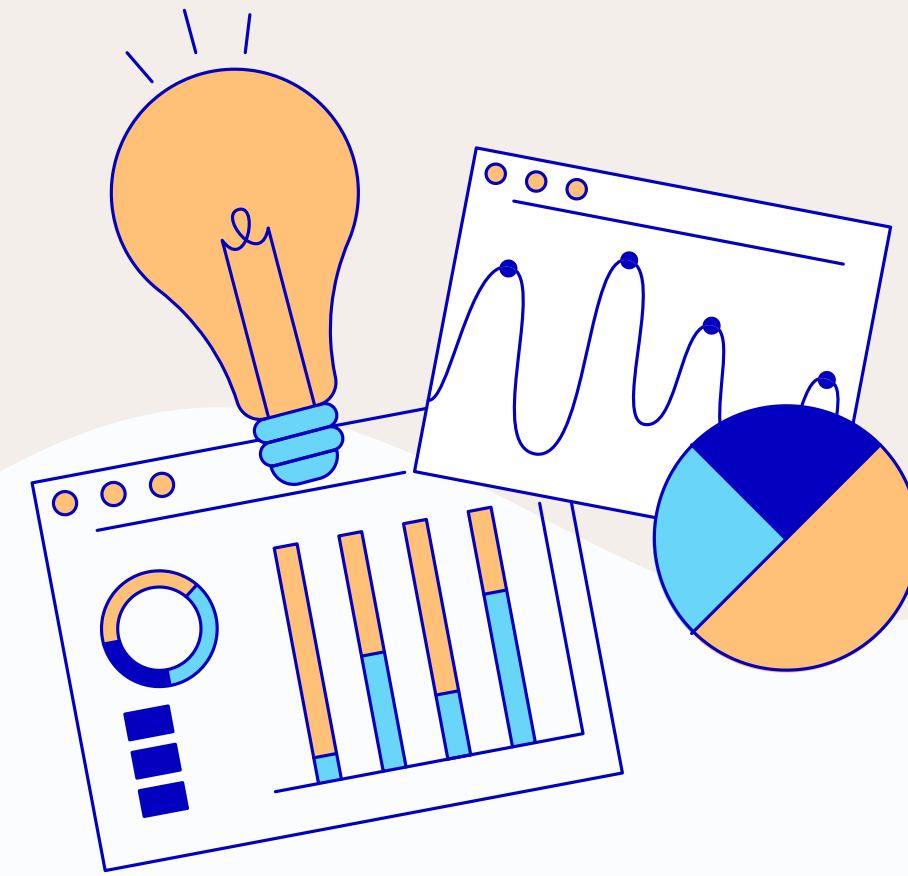


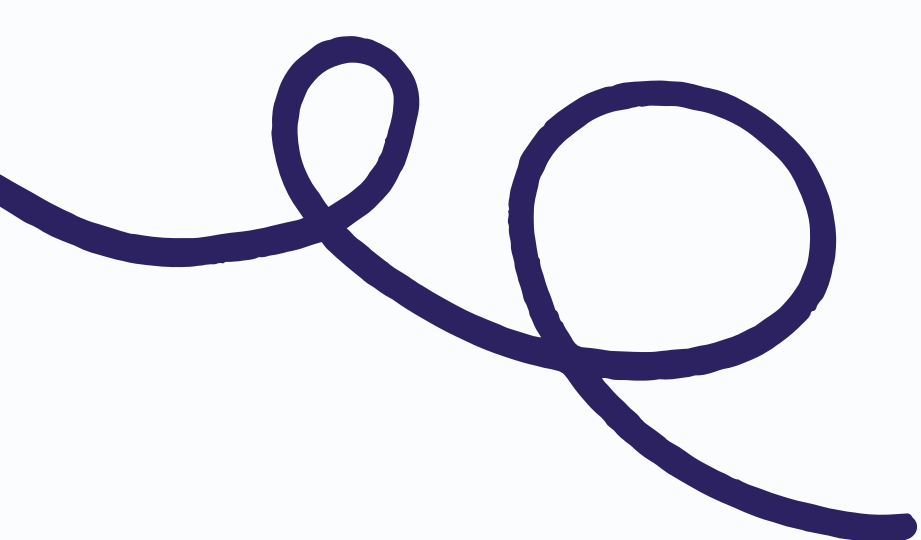
Dataset Details

Identifiers	encounter_id, patient_nbr
Personal Details	race, gender, age, weight
Visit details / identifiers	admission_type_id, discharge_disposition_id, admission_source_id, payer_code object
Treatment / Patient details	num_lab_procedures, num_procedures, num_medications, number_outpatient, number_emergency, number_inpatient
Medicines	'max_glu_serum', 'A1Cresult', 'metformin', 'repaglinide', 'nateglinide', 'chlorpropamide', 'glimepiride', 'acetohexamide', 'glipizide', 'glyburide', 'tolbutamide', 'pioglitazone', 'rosiglitazone', 'acarbose', 'miglitol', 'troglitazone', 'tolazamide', 'examide', 'citoglipton', 'insulin', 'glyburide- metformin', 'glipizide-metformin', 'glimepiride-pioglitazone', 'metformin-rosiglitazone', 'metformin-pioglitazone'
Diagnosis details	diag_1, diag_2 ,diag_3, number_diagnoses, change, diabetesMed
Readmission details	readmitted



DATA CLEANING & PREPROCESSING





Data Cleaning & Preprocessing



01

Handling missing values

Converted all "?" values to proper null values to accurately detect and handle missing data. Used `isnull().sum()` to identify columns with high missing data.

02

Dropped Irrelevant Columns

Removed `weight`, `payer_code`, `encounter_id`, and `patient_nbr` as they were either sparse or non-predictive identifiers.

03

Handled Missing Values in 'race'

Filled missing race values with "Unknown" to avoid data loss and maintain consistency.



04

Mapped ICD Diagnosis Codes for diag_*

Converted raw ICD codes in diagnosis cols into broader disease categories (Circulatory, Respiratory, Diabetes, etc.) to reduce complexity and improve interpretability.



ICD code range	Group
390–459	Circulatory
460–519	Respiratory
520–579	Digestive
250	Diabetes
V-codes	External / Other
Missing	Unknown

05

Filled Missing 'medical_specialty'

Replaced null specialty values with "Unknown" to maintain dataset completeness.

06

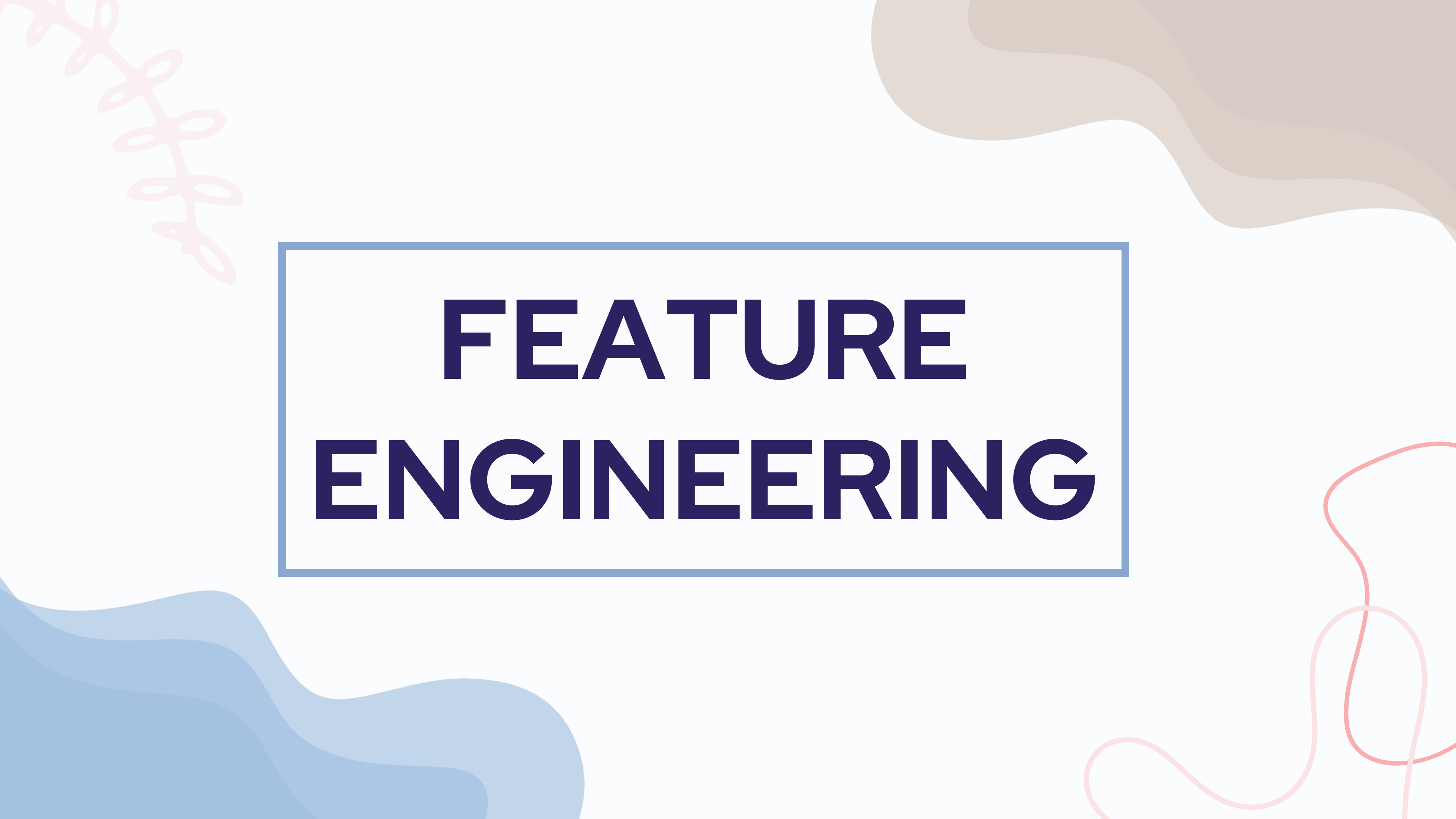
Grouped Medical Specialties

Combined detailed specialties into broader groups (Medicine, Surgery, Cardiology, etc.).

07

Dropped Original Specialty Column

Removed the original medical_specialty column after creating the grouped version to avoid redundancy.



FEATURE ENGINEERING

Feature Engineering

Target Variable Creation

Mapped readmitted into binary target: <30 days = 1 (early readmission), otherwise = 0, and dropped the original column.

Age Column Conversion

Converted age ranges into midpoint numerical values (e.g., [50-60) → 55) to make it usable for modeling.

A1C Result Processing

Norm,>7,>8,Nan

Created two features:

- A1C_measured (whether test was taken)
- A1C_level (Normal/High encoded numerically), then dropped the original column.

Medication Columns Simplification

Dropped 23 individual drug columns since summary variables (diabetesMed, change) already captured medication usage.

Medication Binary Encoding

Converted change and diabetesMed into binary format (Yes/Change = 1, No = 0).



Feature Engineering



Max Glucose Serum Processing

Created `max_glu_serum_measured` and `max_glu_serum_level`, encoded glucose levels numerically, and removed the original column.

Admission Type Grouping

Mapped `admission_type_id` into broader categories (Emergency, Urgent, Elective, etc.) and dropped the original ID column.

Discharge Disposition Handling

Grouped discharge status into categories (Home, Transfer, Expired, Other), removed expired cases, and dropped related columns.

Admission Source Grouping

Mapped `admission_source_id` into meaningful categories (ER, Referral, Transfer, Other) and removed the original column.

Boolean Conversion

Converted all boolean columns into integer format (0/1) to make them compatible with machine learning models.



Feature Re-Engineering

Interaction-Based Risk

- Created Long Stay (≥ 7 days) flag
- Created Elderly (≥ 65 years) flag
- Captures combined hospitalization + age risk
- Removed raw age & stay duration columns

Disease Burden & Complexity

- Combined diagnoses & procedures into disease_burden
- Created complex_case flag (high diagnoses + labs)
- Reflects illness severity & clinical complexity
- Dropped individual diagnosis/lab count columns

Treatment Burden & Instability

- Created polypharmacy flag (≥ 10 meds)
- Created unstable_med (many meds + dosage change)
- Identifies unstable or poorly controlled patients
- Removed raw medication count & change column

Healthcare Utilization Pattern Heading:

- Created total_visits (OP + ER + IP)
- Calculated inpatient_ratio
- Identifies chronic or high-risk patients
- Dropped individual visit-type columns

Diabetes-Focused Indicator

- Created single diabetes_any_diag flag
- Captures diabetes in any diagnosis position
- Simplifies multiple diagnosis columns
- Reduces dimensionality

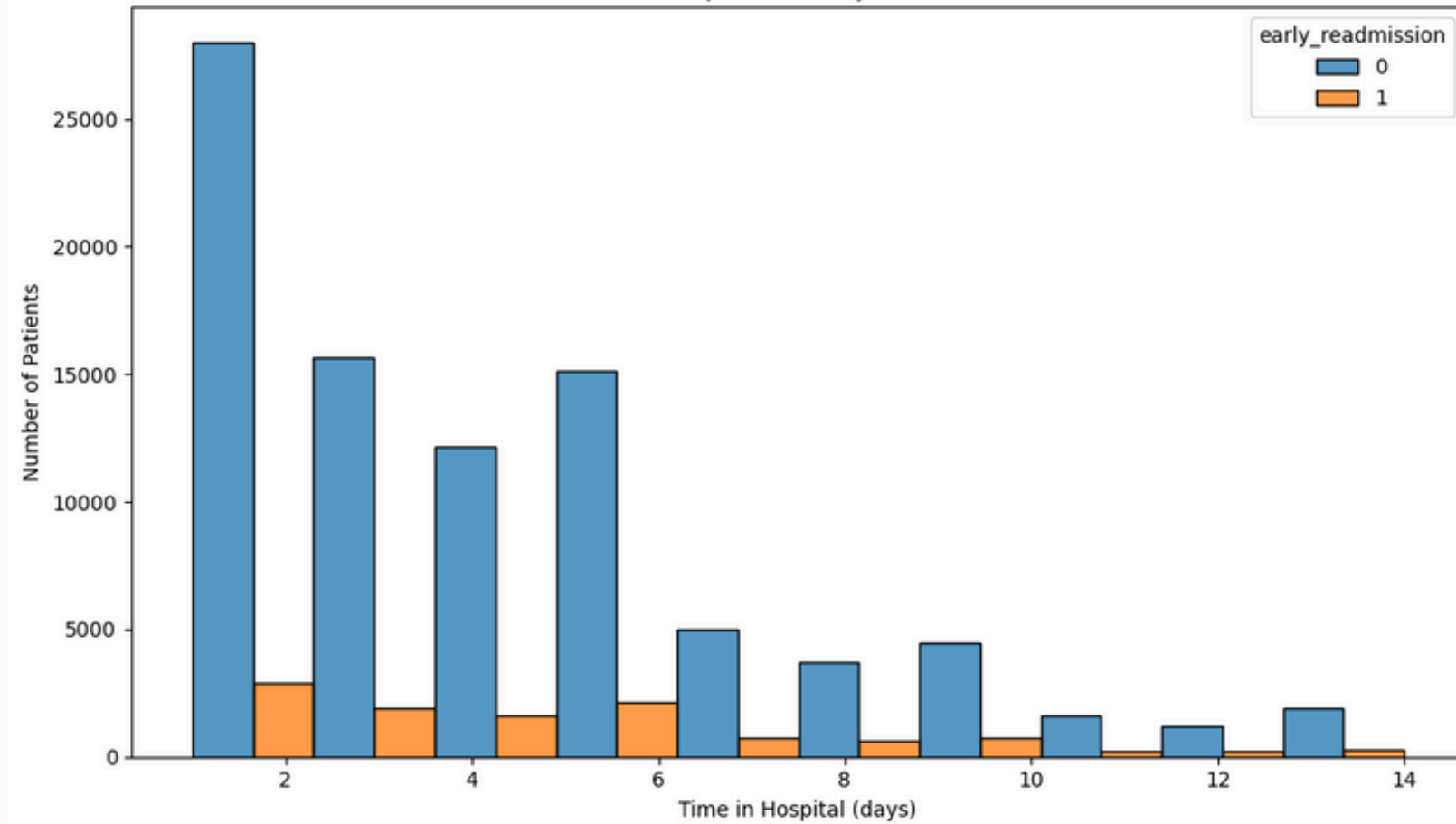


Exploratory Data Analysis

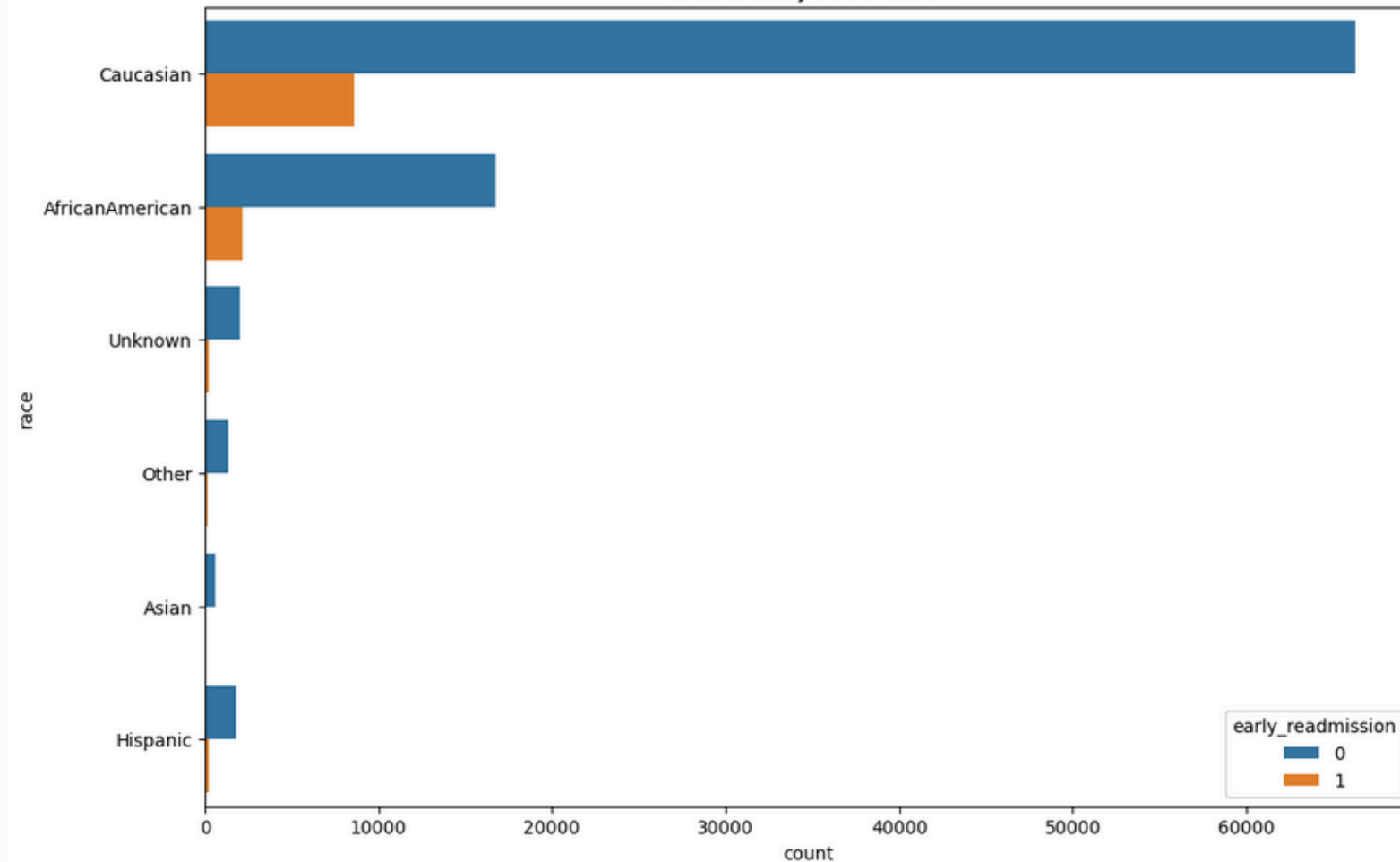


DATA VISUALIZATION

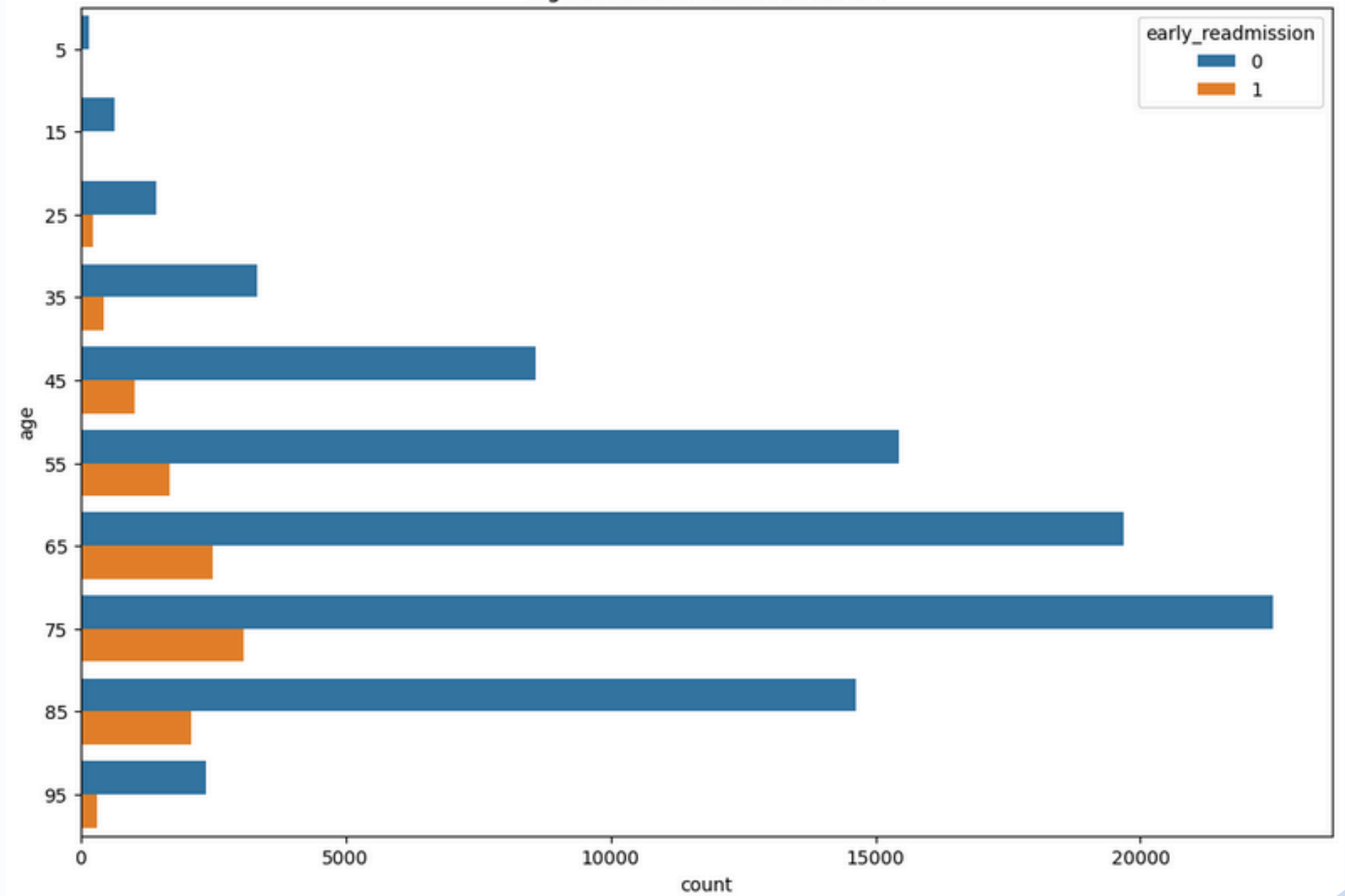
Time in hospital VS Early Readmission

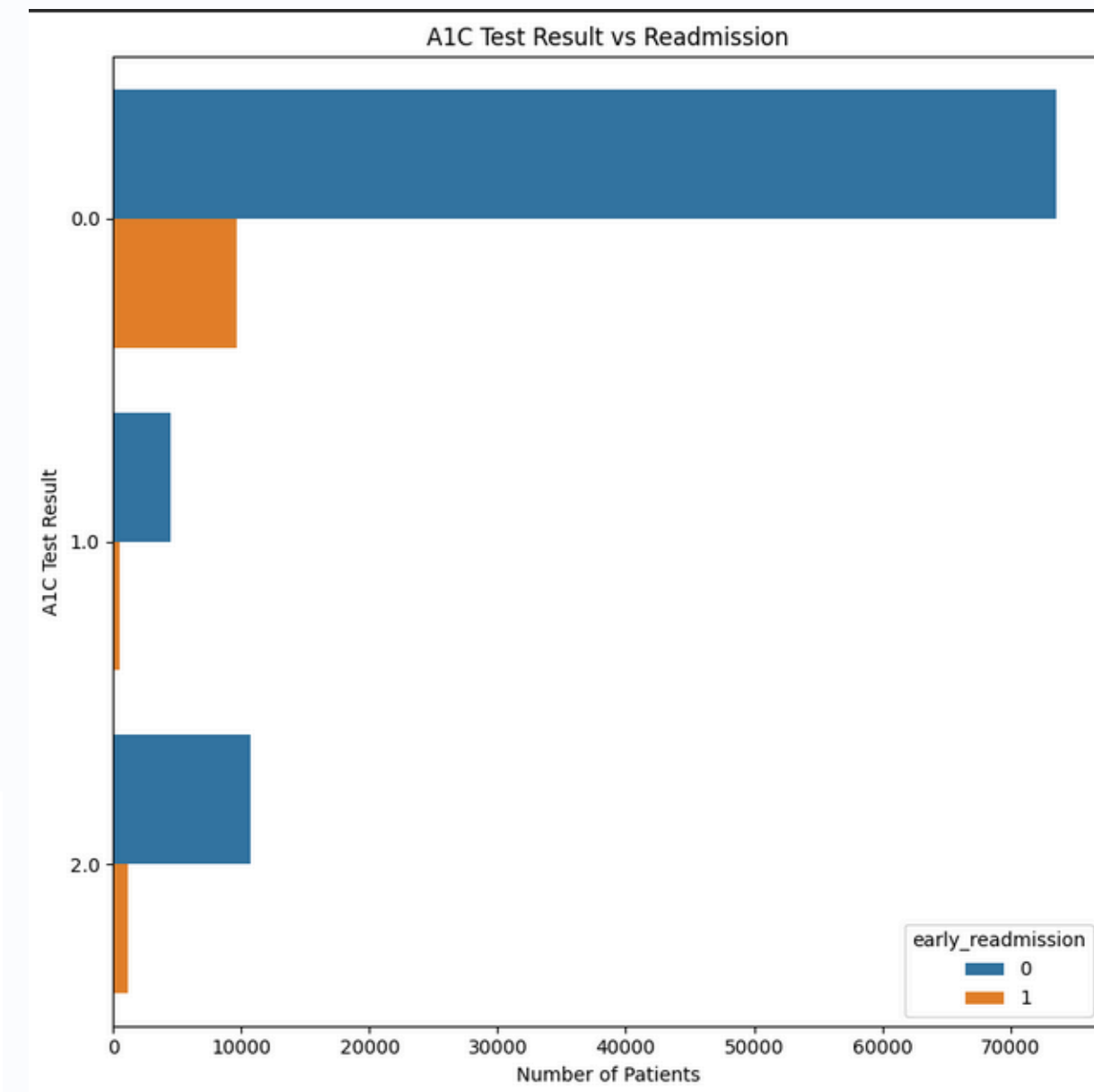
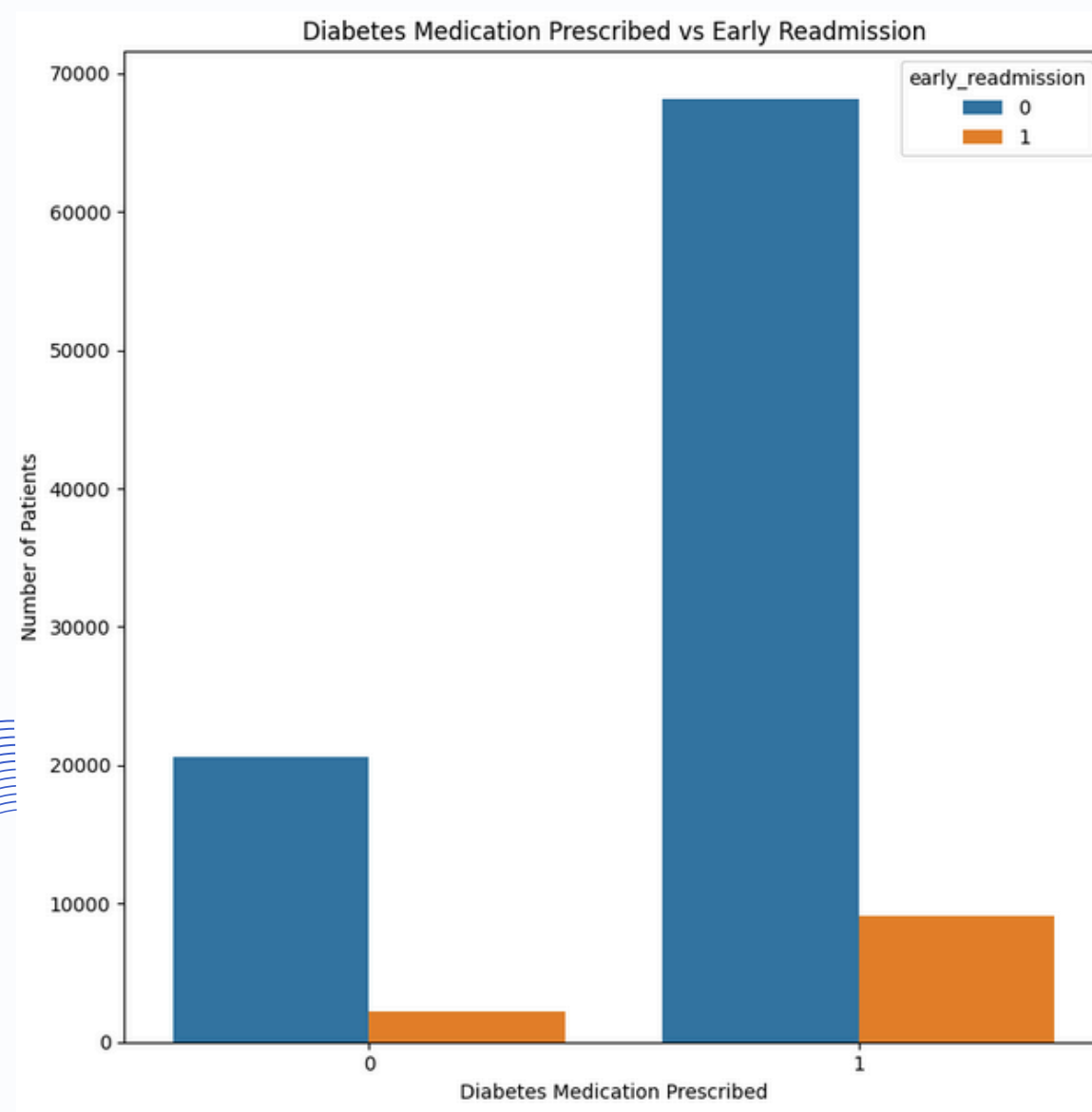
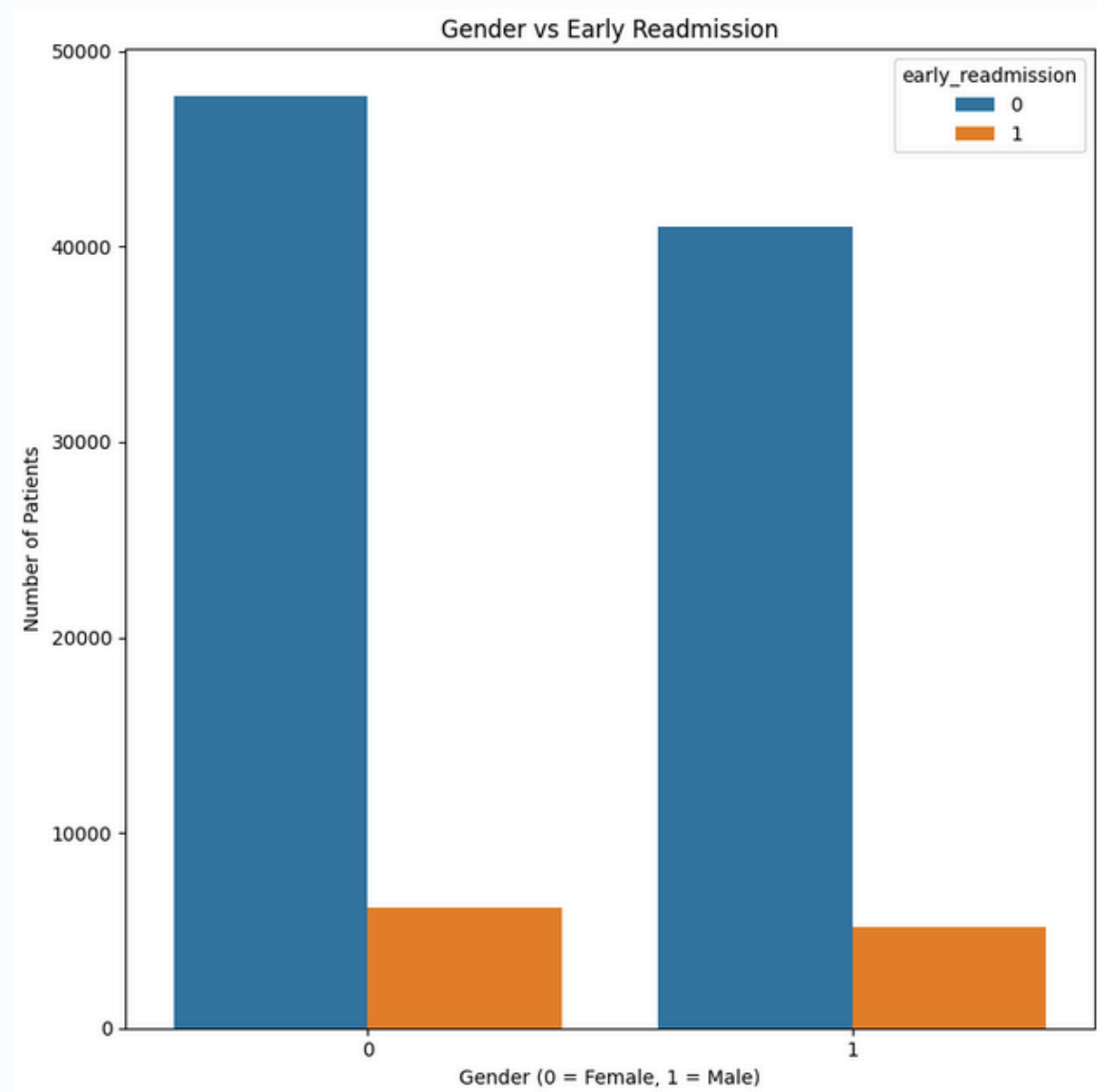


Race vs Early Readmission



Age of Patient VS. Readmission





OUTLIER DETECTION

1. The Interquartile Range (IQR)

$$\text{IQR} = Q3 - Q1$$

2. The "Whiskers" (Tukey's Fences)

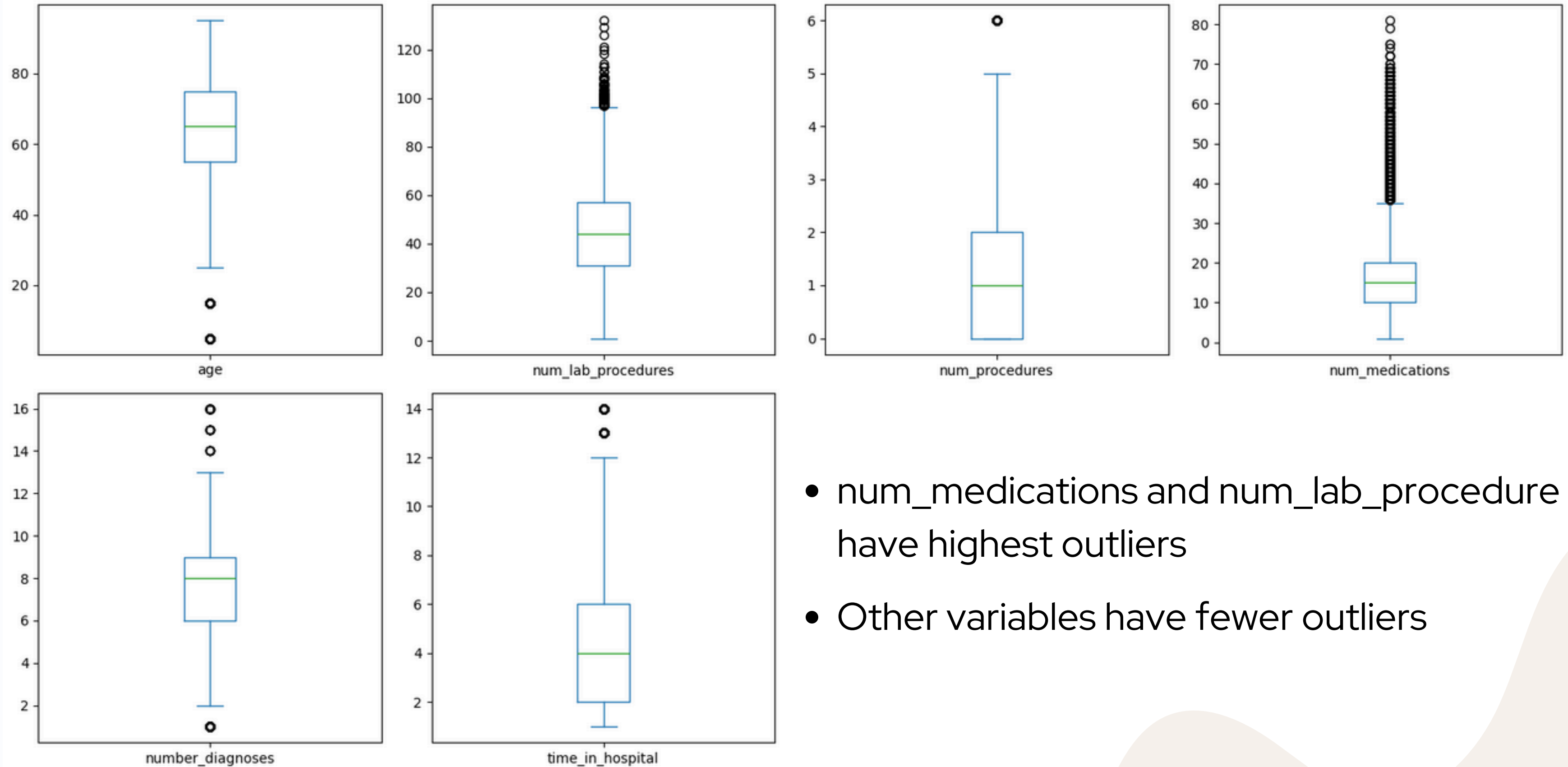
Lower Whisker Bound: $Q1 - 1.5 \times \text{IQR}$

Upper Whisker Bound: $Q3 + 1.5 \times \text{IQR}$

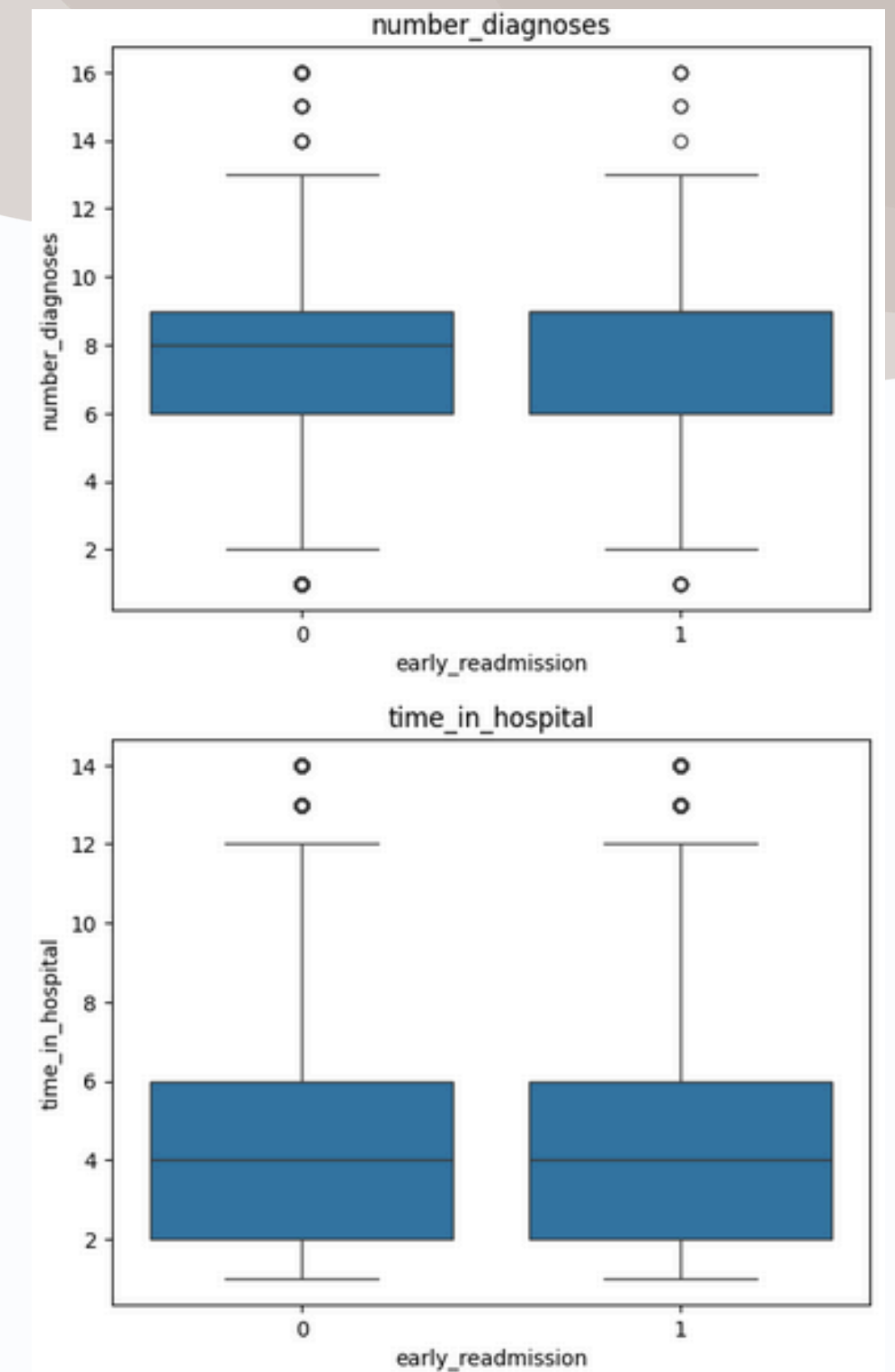
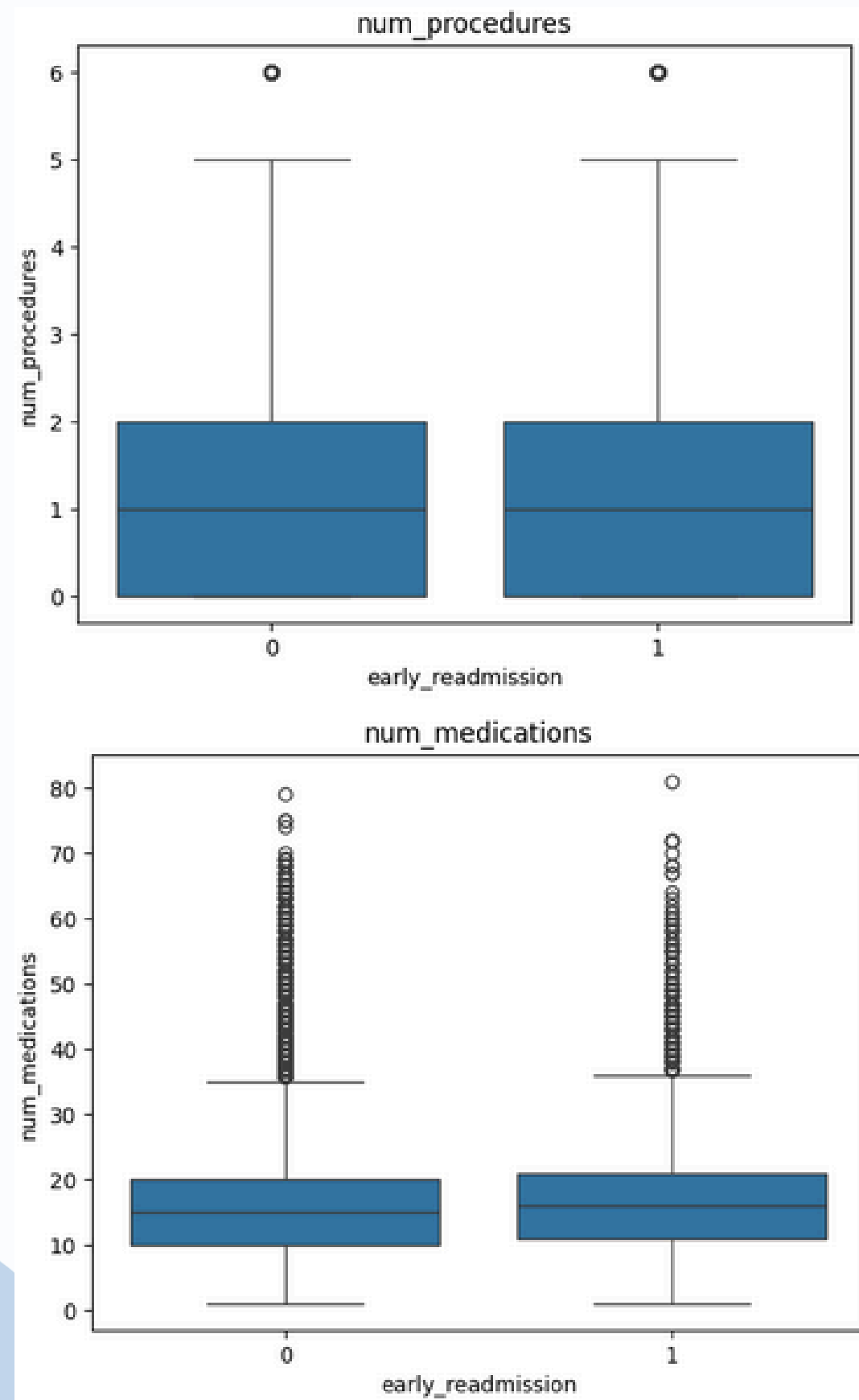
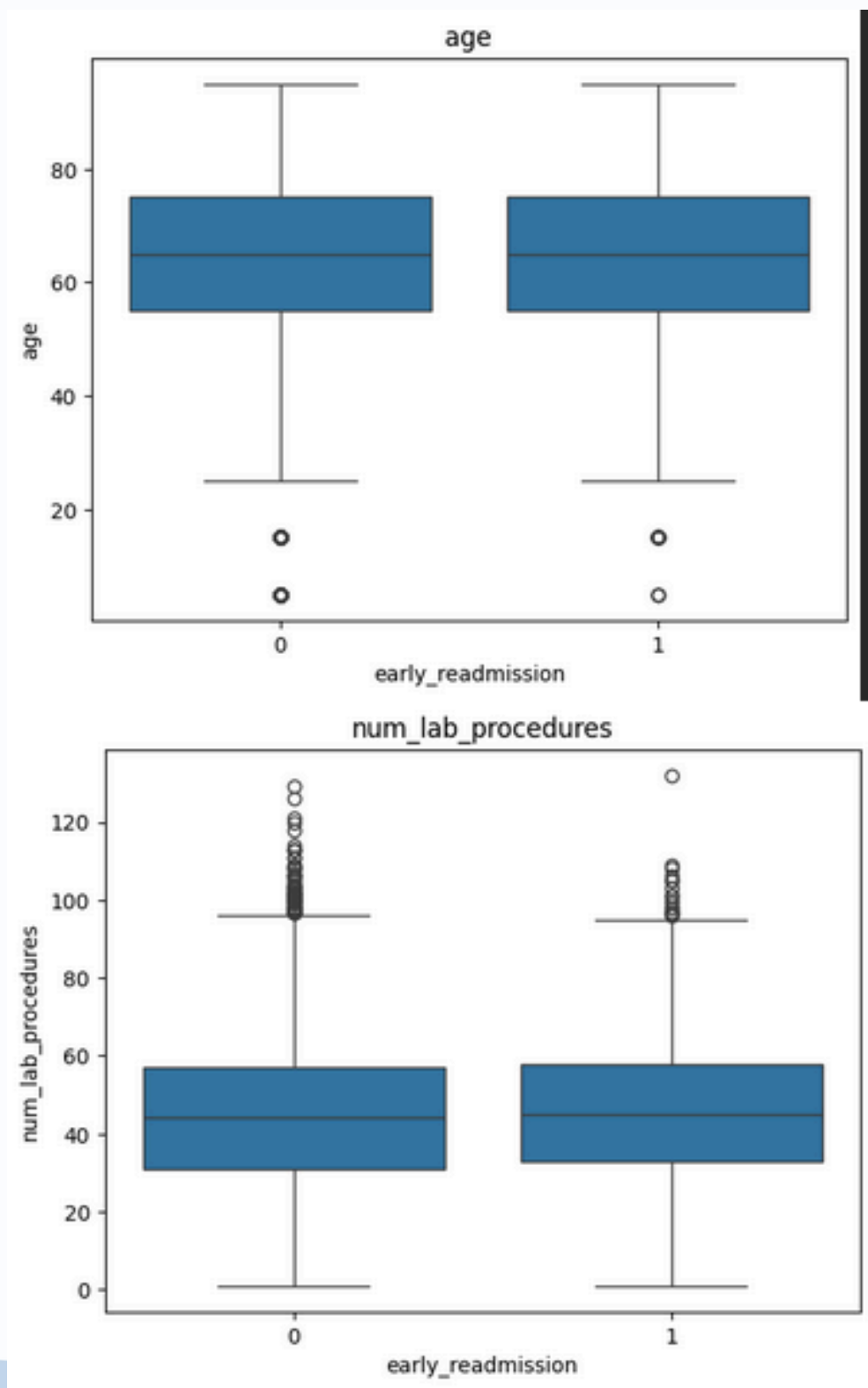
3. Outlier Detection

$$\text{Outlier} > Q3 + 1.5 \times \text{IQR} \text{ or } \text{Outlier} < Q1 - 1.5 \times \text{IQR}$$

OUTLIER DETECTION



- num_medications and num_lab_procedure have highest outliers
- Other variables have fewer outliers

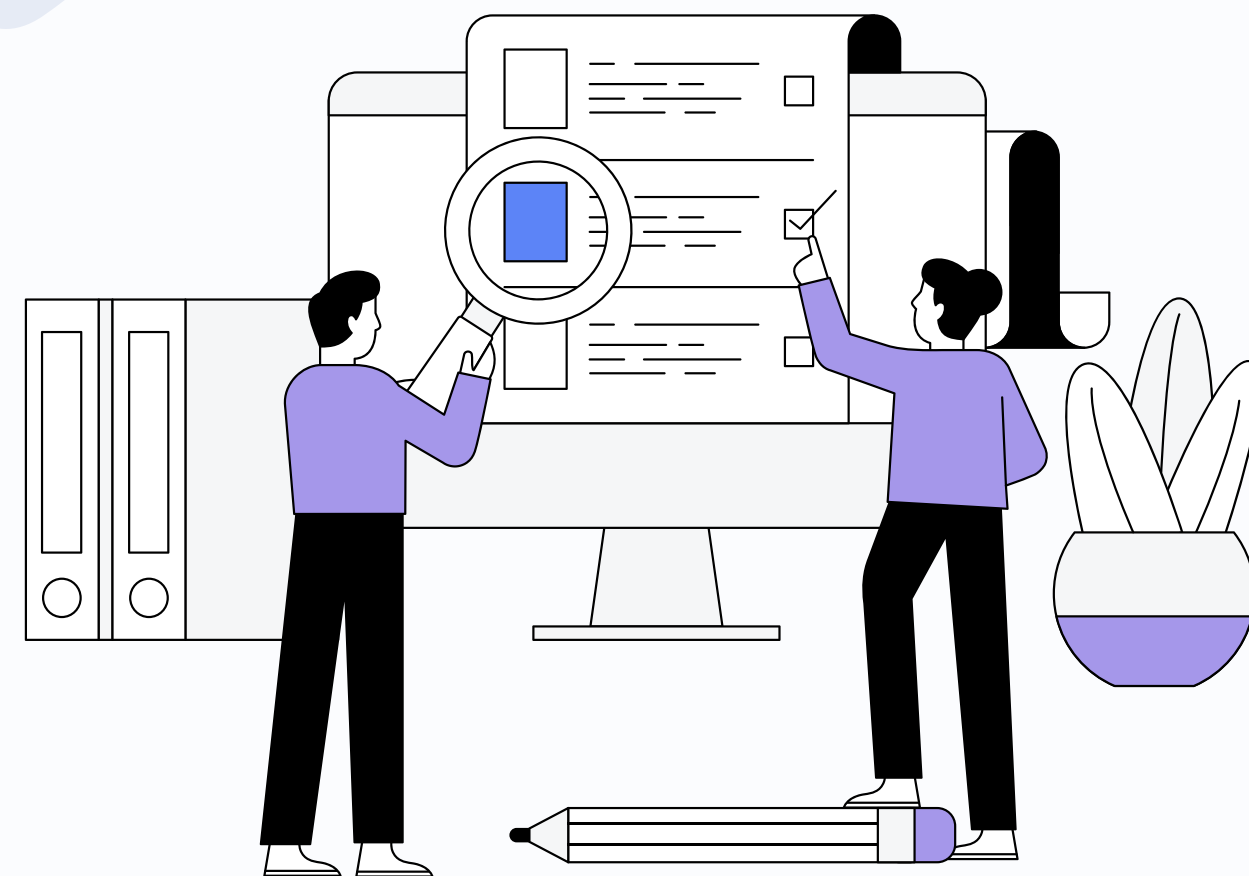


OUTLIER HANDLING

- Outlier Strategy & Implementation
- Technique: IQR-Based Capping (Clipping).
- **`df[col].clip(lower=lower_bound, upper=upper_bound)`**
- Strategic Exception: Intentionally preserved outliers in Age and Diagnosis Counts.
- Data Integrity: Confirmed by `df.shape`; we reduced noise while maintaining 100% of our data records



ASSUMPTION CHECKING



ASSUMPTION CHECKING

1. Linearity with Log-Odds:

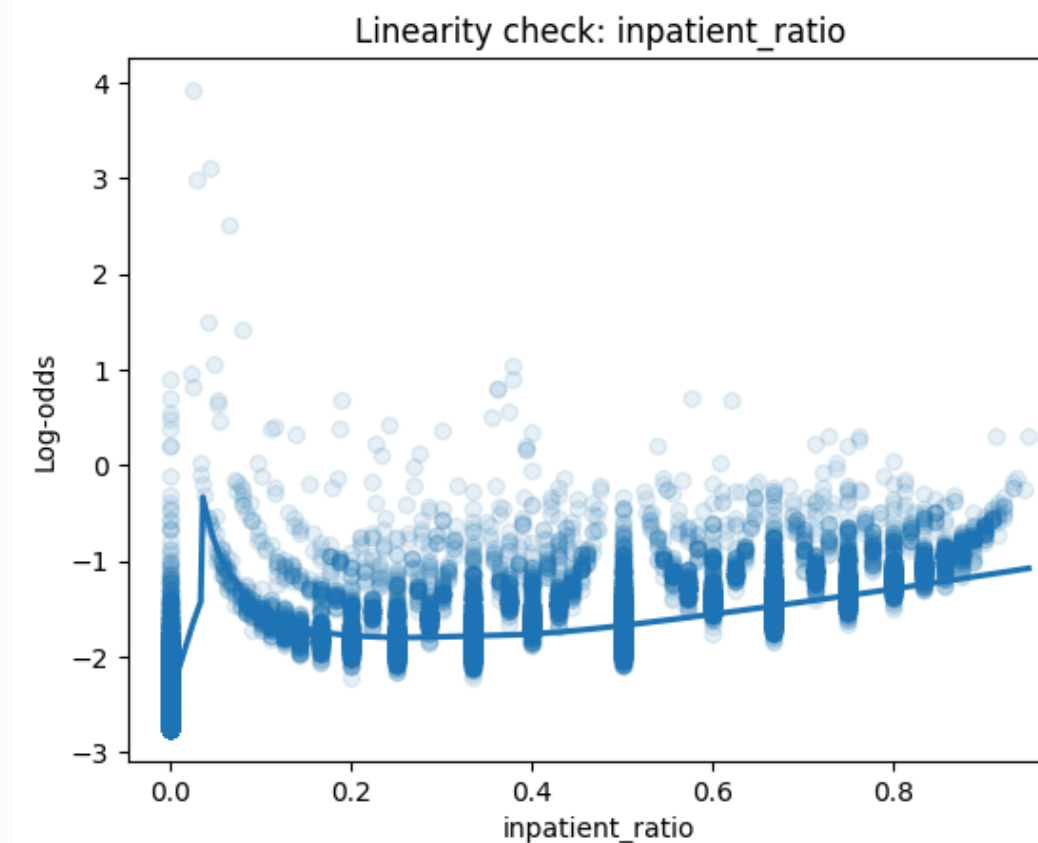
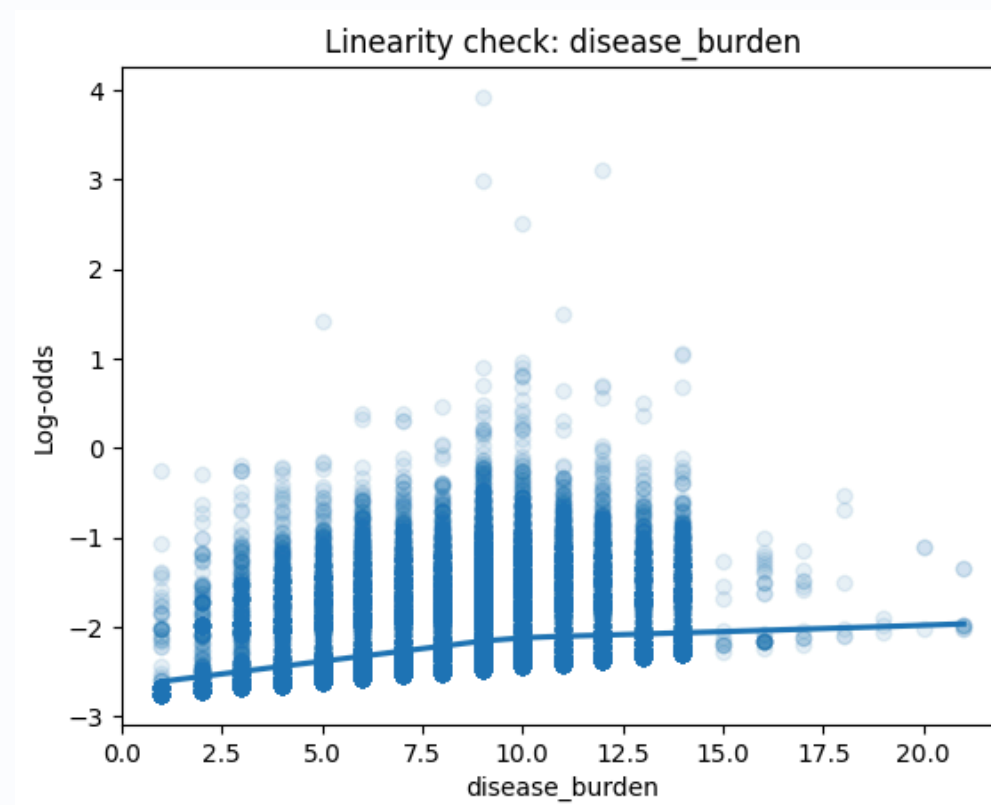
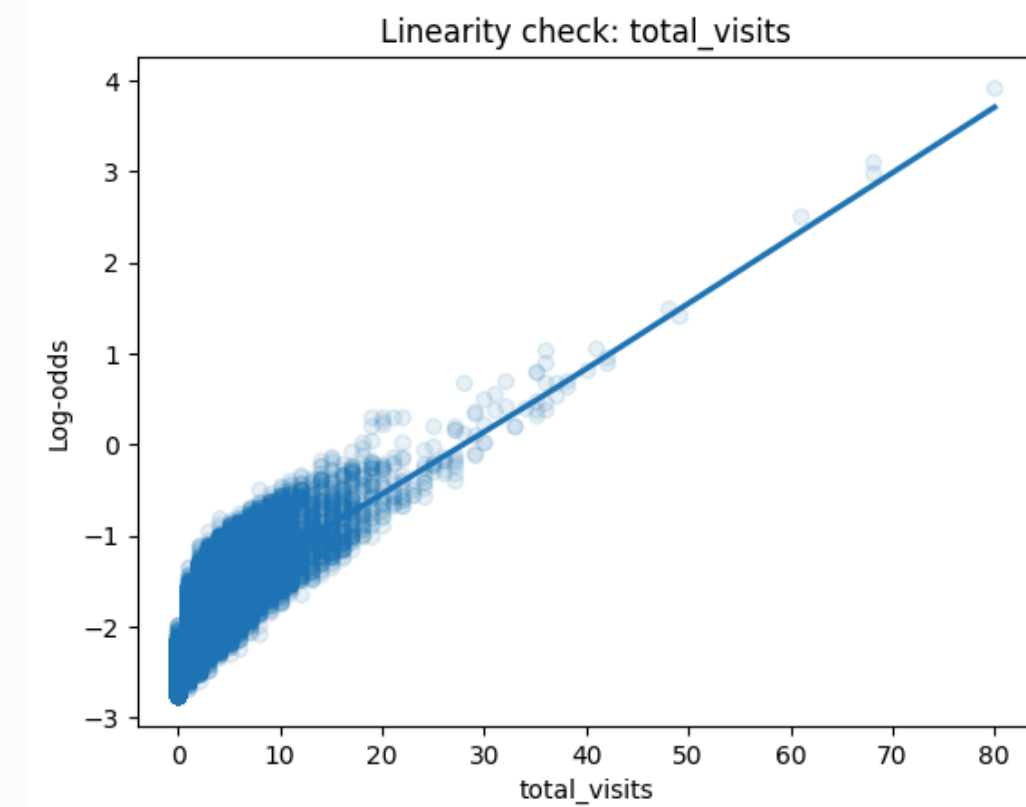
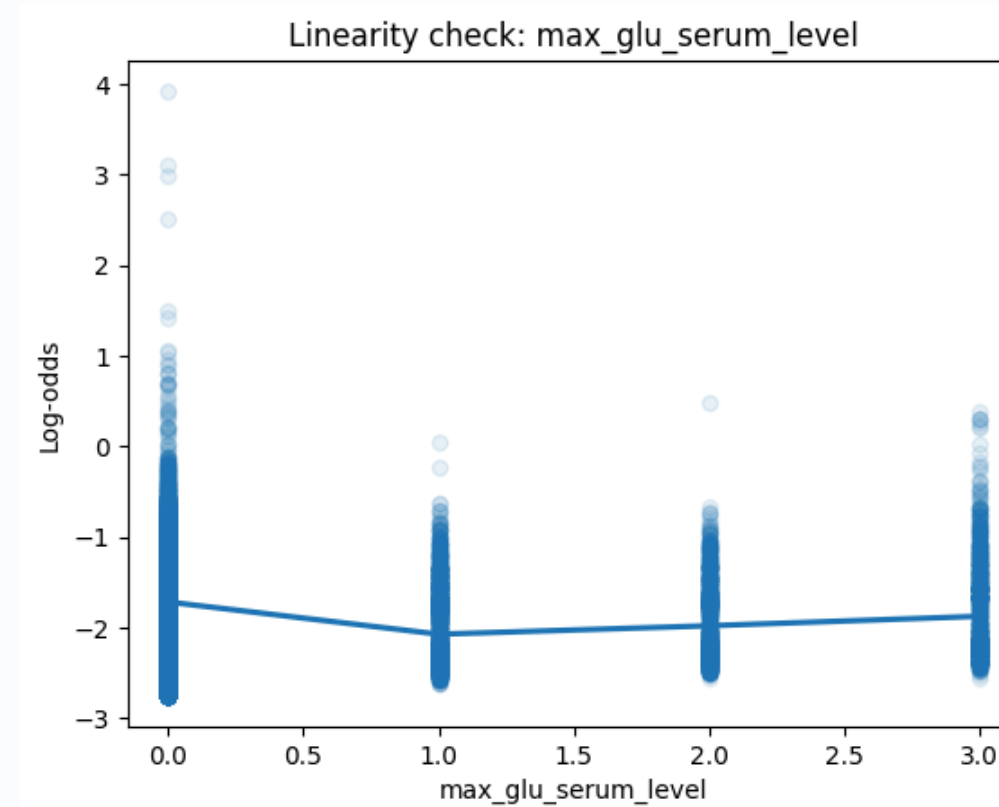
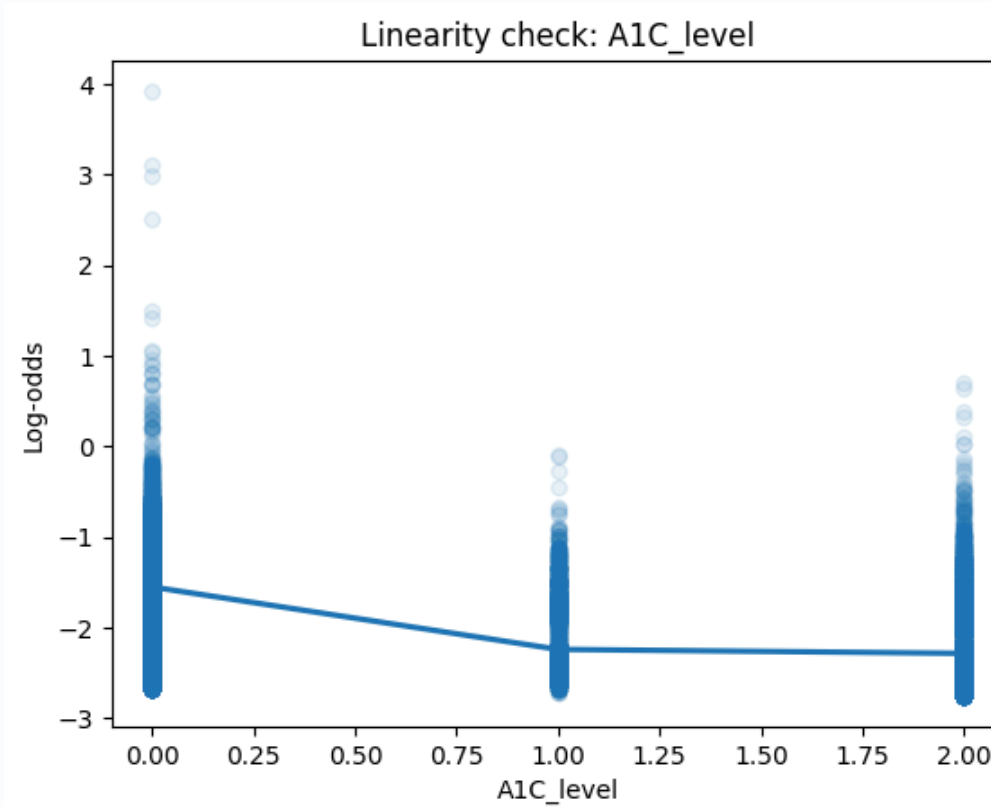
Graph: Log-Odds vs Predictors (A1C_level, max_glu_serum_level, total_visits, inpatient_ratio)

- Log-odds plotted against continuous predictors
- Relationship appears approximately linear
- No strong curvature observed

Conclusion: Linearity of log-odds satisfied

ASSUMPTION CHECKING

1. Linearity with Log-Odds:



```
=== Linearity check: Coefficients ===
const                -2.705909
A1C_level             -0.040236
max_glu_serum_level   0.065245
disease_burden        0.034044
total_visits          0.078660
inpatient_ratio       1.134975
dtype: float64
```

ASSUMPTION CHECKING

2. Multicollinearity:

- Occurs when independent variables are highly correlated.
- Leads to unstable coefficients, inflated standard errors, and unreliable p-values.

Multicollinearity can be checked using VIF

$$VIF\ x_i = \frac{1}{Tolerance} = \frac{1}{1 - R_i^2}$$

- $VIF \approx 1 \rightarrow$ No multicollinearity
- $1 < VIF \leq 5 \rightarrow$ Moderate correlation (acceptable)
- $VIF > 5 \rightarrow$ High multicollinearity
- $VIF > 10 \rightarrow$ Serious multicollinearity

ASSUMPTION CHECKING

2. Multicollinearity:

	Feature	VIF
4	inpatient_ratio	1.755975
2	disease_burden	1.689308
3	total_visits	1.588712
0	A1C_level	1.166328
1	max_glu_serum_level	1.039235

- $VIF < 5 \rightarrow$ Low multicollinearity
- No variable crosses the critical limit
- Conclusion: Multicollinearity not present $\rightarrow$ Model stable

ASSUMPTION CHECKING

2. Multicollinearity: checking correlation (diagnostic).

Feature 1
A1C_level
max_glu_serum_measured
admission_source_grouped_Transfer
admission_source_grouped_Physician_Referral
medical_specialty_grouped_Unknown
admission_source_grouped_Transfer
admission_type_grouped_Urgent
admission_type_grouped_Other
admission_source_grouped_Transfer
admission_source_grouped_Transfer
unstable_med
inpatient_ratio
admission_type_grouped_Other
unstable_med
diag_1_Diabetes
diabetes_any_diag
polypharmacy
medical_specialty_grouped_Unknown
admission_source_grouped_Physician_Referral
medical_specialty_grouped_Unknown

Feature 2	Correlation
A1C_measured	0.959828
max_glu_serum_level	0.902054
max_glu_serum_measured	0.620597
admission_type_grouped_Emergency	-0.608346
medical_specialty_grouped_Medicine	-0.555067
max_glu_serum_level	0.531169
admission_type_grouped_Emergency	-0.499245
max_glu_serum_measured	0.496762
admission_type_grouped_Newborn	0.465113
admission_type_grouped_Other	0.464448
diabetesMed	0.451872
total_visits	0.439792
max_glu_serum_level	0.433974
polypharmacy	0.418012
diabetes_any_diag	0.396265
disease_burden	-0.349497
disease_burden	0.310288
medical_specialty_grouped_Other	-0.292461
medical_specialty_grouped_Other	0.280612
medical_specialty_grouped_Emergency	-0.278443

- there is high relation between measured and level variable as one encodes the other, that is level only has value if the test was measured first (1).
- so we drop measured and keep level variable as its more meaningful

ASSUMPTION CHECKING

3. Binary Target Variable:

- Target variable is encoded into two classes (0 and 1)
- Logistic regression requires binary outcome

```
Unique Values in Target Column after Mapping: [0 1]
early_readmission
0      90409
1      11357
Name: count, dtype: int64
```

Conclusion: Binary target assumption satisfied

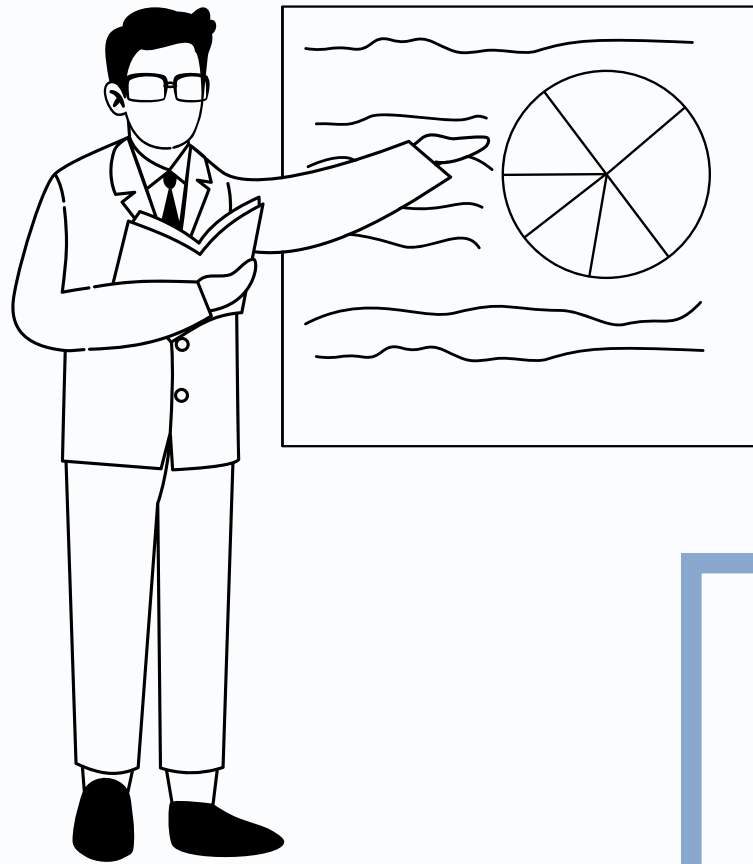
ASSUMPTION CHECKING

4. Sufficient Sample Size:

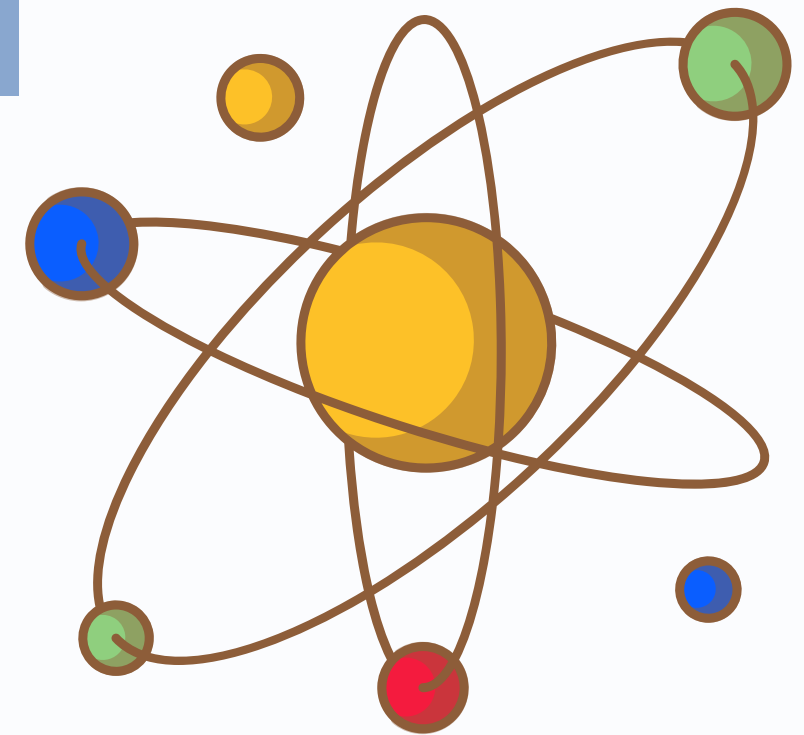
- Adequate observations per predictor
- Ensures stable and reliable estimates
- EPV $\gg 10$ (recommended threshold for logistic regression)
- EPV = 299 $\Rightarrow$ Large dataset used for modeling

```
Total early admissions 11357  
Total predictors (X cols): 38  
Events per variable 298.86842105263156
```

Conclusion: Sample size sufficient



HYPOTHESIS TESTING



Hypothesis Testing

Assessing Statistical Significance of Predictors

Objective: Check which clinical and demographic factors are statistically associated with early readmission (≤ 30 days).

Null Hypothesis (H_0): No predictor has a significant effect (all $\beta_i = 0$).

Alternative Hypothesis (H_1): At least one predictor significantly affects early readmission ($\beta_i \neq 0$).

Method: Logistic regression on full dataset (before train/test split).

Hypothesis Testing

Optimization terminated successfully.
Current function value: 0.341151
Iterations 7

Logit Regression Results

Dep. Variable:	early_readmission	No. Observations:	100121
Model:	Logit	Df Residuals:	100002
Method:	MLE	Df Model:	38
Date:	Thu, 12 Feb 2026	Pseudo R-squ.:	0.03530
Time:	20:38:35	Log-Likelihood:	-34156.
converged:	True	LL-Null:	-35406.
Covariance Type:	nonrobust	LLR p-value:	0.000

	coef	std err	z	P> z	[0.025	0.975]
const	-3.2724	0.091	-36.086	0.000	-3.450	-3.095
gender	0.0072	0.020	0.352	0.724	-0.033	0.047
diabetesMed	0.1448	0.028	5.116	0.000	0.089	0.200
A1C_measured	-0.1231	0.030	-4.168	0.000	-0.181	-0.065
max_glu_serum_measured	0.2312	0.059	3.908	0.000	0.115	0.347
long_stay	0.1086	0.026	4.242	0.000	0.058	0.159
elderly	0.1564	0.023	6.709	0.000	0.111	0.202
disease_burden	0.0231	0.005	4.872	0.000	0.014	0.032
complex_case	0.0574	0.023	2.501	0.012	0.012	0.102
polypharmacy	0.1121	0.031	3.567	0.000	0.050	0.174
unstable_med	0.0112	0.025	0.456	0.649	-0.037	0.060
total_visits	0.0784	0.004	20.285	0.000	0.071	0.086
inpatient_ratio	1.0596	0.039	27.096	0.000	0.983	1.136
diabetes_any_diag	0.0386	0.025	1.553	0.120	-0.010	0.087
race_Asian	-0.0036	0.135	-0.027	0.979	-0.268	0.261
race_Caucasian	0.0020	0.027	0.074	0.941	-0.051	0.055
race_Hispanic	-0.0498	0.078	-0.642	0.521	-0.202	0.102
race_Other	-0.0779	0.092	-0.848	0.396	-0.258	0.102
race_Unknown	-0.1770	0.081	-2.186	0.029	-0.336	-0.018
medical_specialty_grouped_Emergency	0.2931	0.069	4.269	0.000	0.159	0.428
medical_specialty_grouped_Medicine	0.3773	0.057	6.642	0.000	0.266	0.489
medical_specialty_grouped_Other	0.2690	0.065	4.111	0.000	0.141	0.397
medical_specialty_grouped_Pediatrics	-0.4493	0.223	-2.017	0.044	-0.886	-0.013
medical_specialty_grouped_Psychiatry	0.4807	0.117	4.122	0.000	0.252	0.709
medical_specialty_grouped_Surgery	0.3170	0.073	4.359	0.000	0.174	0.460
medical_specialty_grouped_Unknown	0.3577	0.055	6.504	0.000	0.250	0.465
diag_1_Diabetes	0.0690	0.042	1.642	0.100	-0.013	0.151
diag_1_Digestive	-0.1487	0.040	-3.739	0.000	-0.227	-0.071
diag_1_External/Other	0.1221	0.074	1.641	0.101	-0.024	0.268
diag_1_Other	-0.0892	0.025	-3.505	0.000	-0.139	-0.039
diag_1_Respiratory	-0.1815	0.038	-4.781	0.000	-0.256	-0.107
diag_1_Unknown	0.7339	0.519	1.414	0.157	-0.283	1.751
admission_type_grouped_Emergency	0.0658	0.045	1.465	0.143	-0.022	0.154
admission_type_grouped_Newborn	0.0702	0.064	1.089	0.276	-0.056	0.196
admission_type_grouped_Other	-0.0807	0.066	-1.215	0.224	-0.211	0.050
admission_type_grouped_Urgent	0.0589	0.037	1.604	0.109	-0.013	0.131
admission_source_grouped_Other	-0.0902	0.046	-1.976	0.048	-0.180	-0.001
admission_source_grouped_Physician_Referral	0.0073	0.040	0.184	0.854	-0.071	0.085
admission_source_grouped_Transfer	-0.1469	0.064	-2.309	0.021	-0.272	-0.022

(Likelihood Ratio Test) LLR p-value = 0 < 0.05

- predictors collectively improve model compared to a null model with no predictors. Thus, we reject the null hypothesis

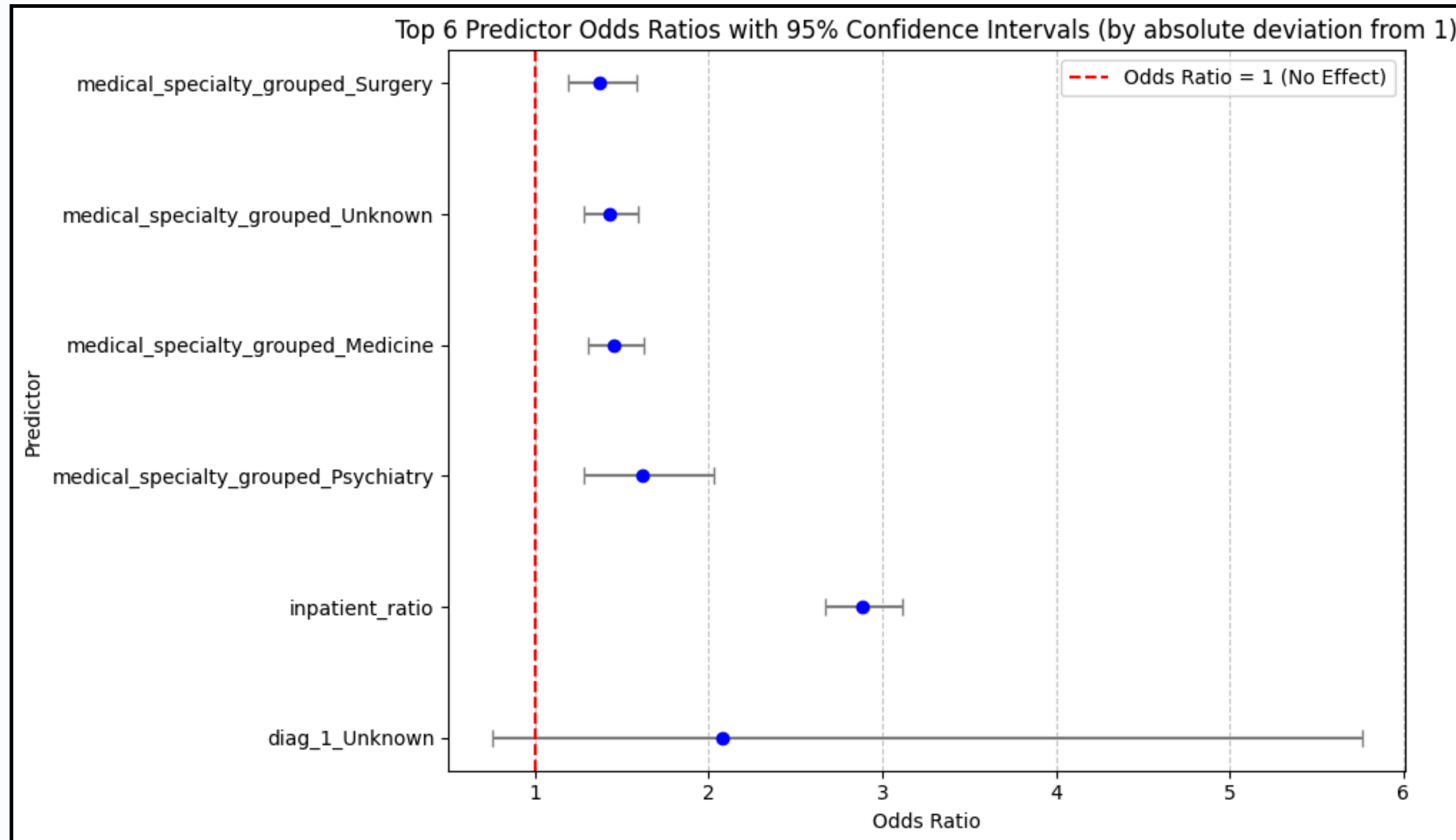
Log-Likelihood: -34156 > LL-Null: -35406.

- Full model improved performance

Pseudo R-squ.: 0.03530 = 3.5% improvement

- modest explanatory power, but expected for clinical readmission models where outcomes depend on multifactorial influences beyond recorded variables

Hypothesis Testing

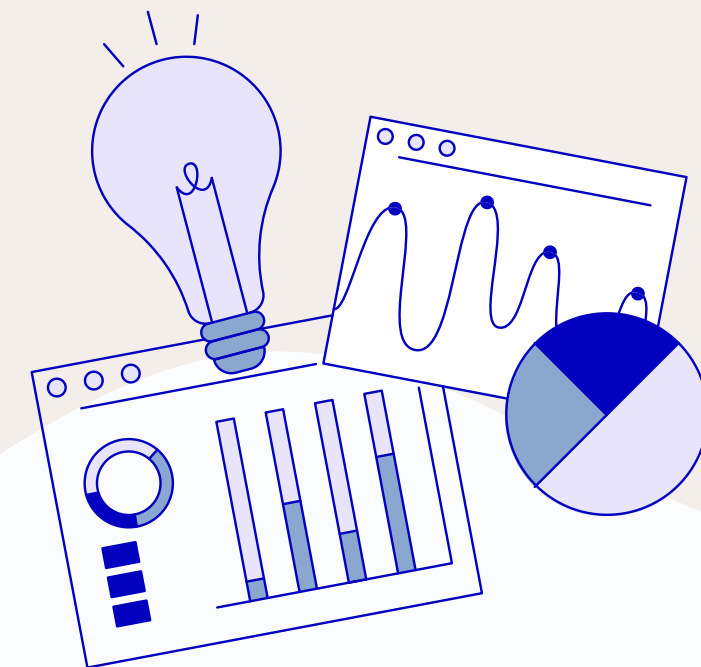


Forest plot of top 6 predictors with ORs and 95% CI.

- Red vertical line = OR = 1 (no effect)
- $OR > 1 \rightarrow$ increases odds of readmission
- $OR < 1 \rightarrow$ decreases odds
- CIs crossing 1 $\rightarrow$ not significant

Most top predictors' CIs do not cross 1 $\rightarrow$ statistically significant.

$\Rightarrow$ Confirms that logistic regression is appropriate for modeling risk.



MODELLING



Logistic Regression:

Logistic Regression is a supervised learning algorithm used primarily for binary classification tasks, where the outcome is categorical and has two possible labels. It estimates the probability that a given instance belongs to a specific class.

Logit Function :

$$\ln \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$$

Where:

- $p = P(Y = 1 | X) \rightarrow$ Probability of positive class
- $1 - p = P(Y = 0 | X) \rightarrow$ Probability of negative class
- $\frac{p}{1-p} \rightarrow$ Odds
- $\ln \left(\frac{p}{1-p} \right) \rightarrow$ Log-odds (Logit)
- $\beta_0 \rightarrow$ Intercept (bias term)
- $\beta_i \rightarrow$ Coefficient of feature X_i
- $X_i \rightarrow$ Input features

- Here we find the intercept and coefficients such that MLE is maximized

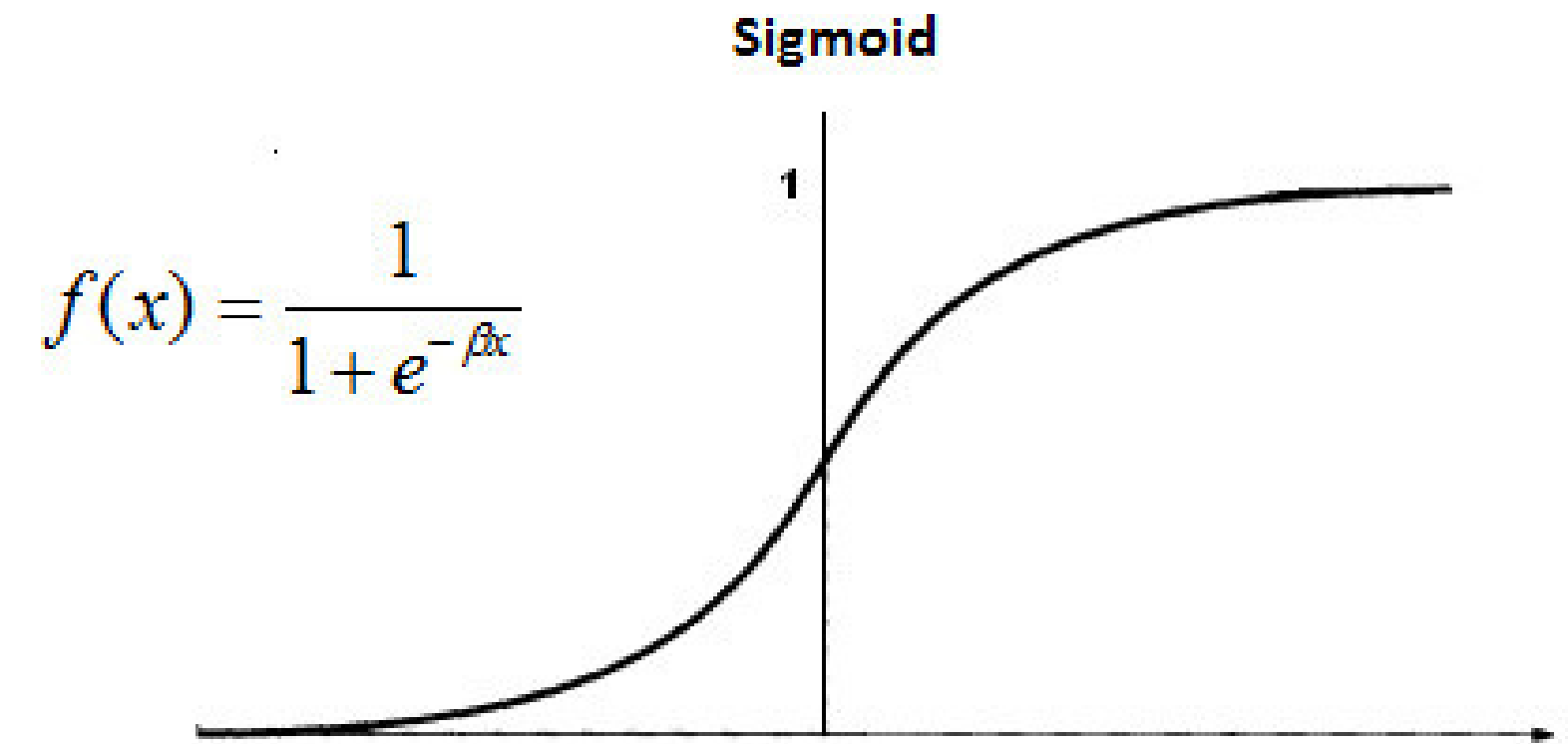
```
from sklearn.linear_model import LogisticRegression
model = LogisticRegression()
model.fit(X_train, y_train)
```

Sigmoid (Logistic) Function

- To convert the linear output z into probability:

$$p = \frac{1}{1 + e^{-z}}$$

`model.predict_proba()`



Binary Classification Decision:

- Default Threshold (0.5)
- If $p \geq 0.5 \rightarrow$ Class 1
else: $\rightarrow$ Class 0

Optimal Probability Threshold $P(P^*)$:

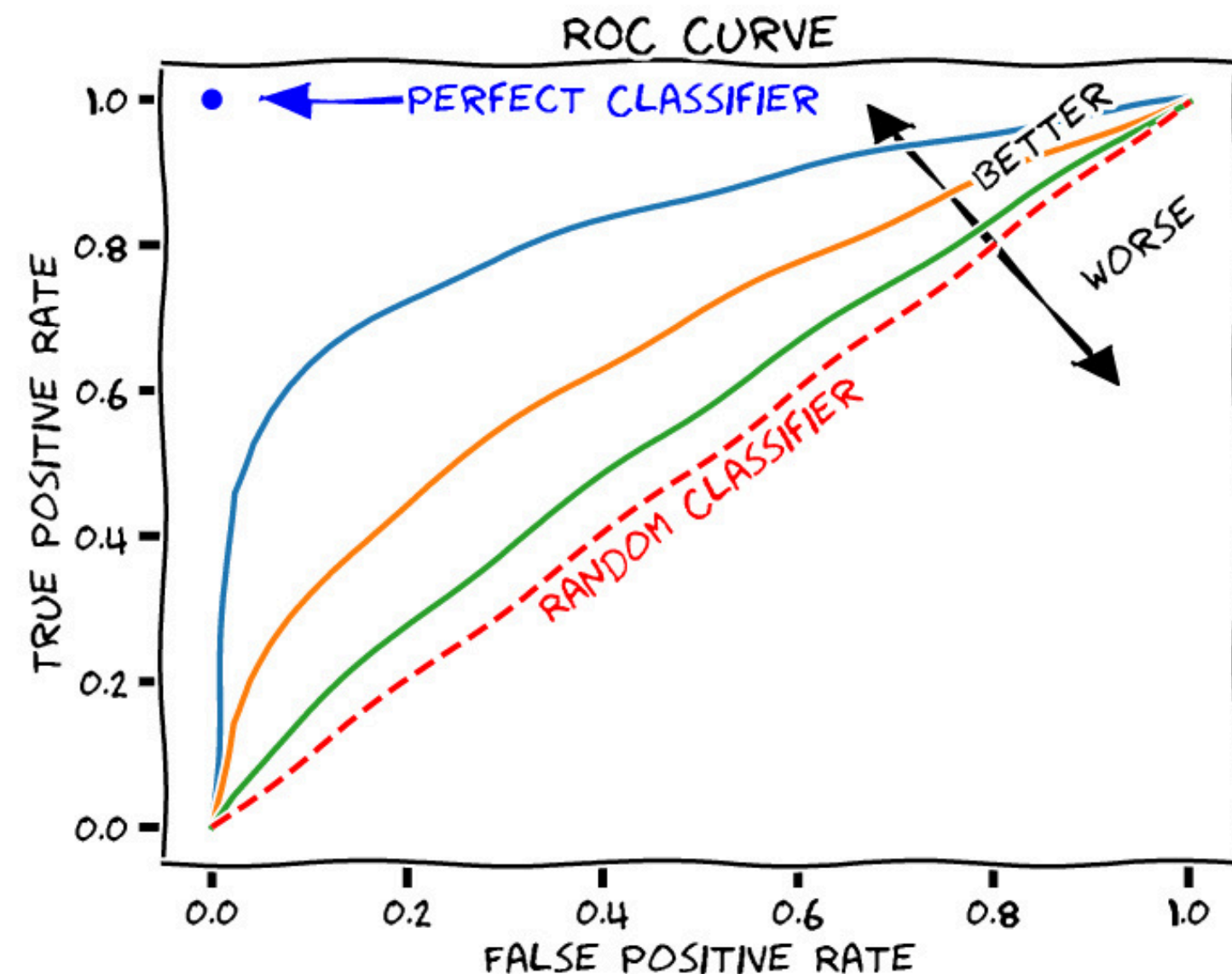
Instead of using 0.5, we can find an optimal threshold using ROC curve:

$$P^* = \text{Threshold minimizing } \sqrt{(FPR)^2 + (1 - TPR)^2}$$

FPR $\rightarrow$ False Positive Rate $\rightarrow$ (1-Specificity)

TPR $\rightarrow$ True Positive Rate $\rightarrow$ Sensitivity

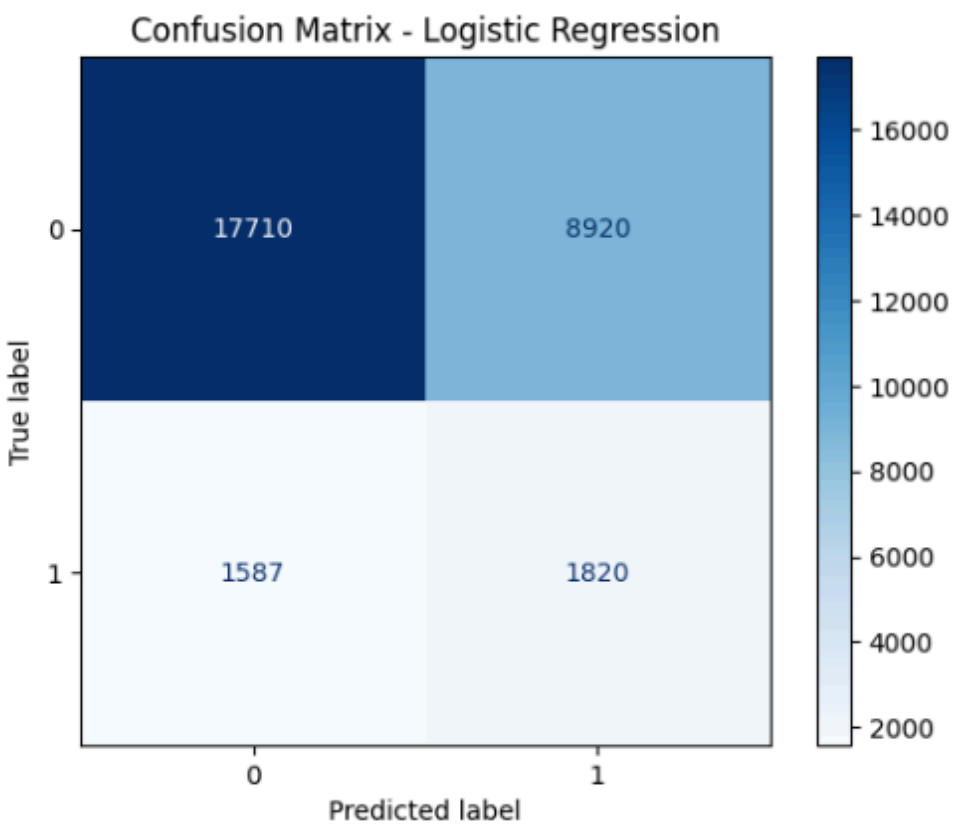
We find P^* in such a way that Sensitivity and (1-Specificity) are balanced



Outcomes

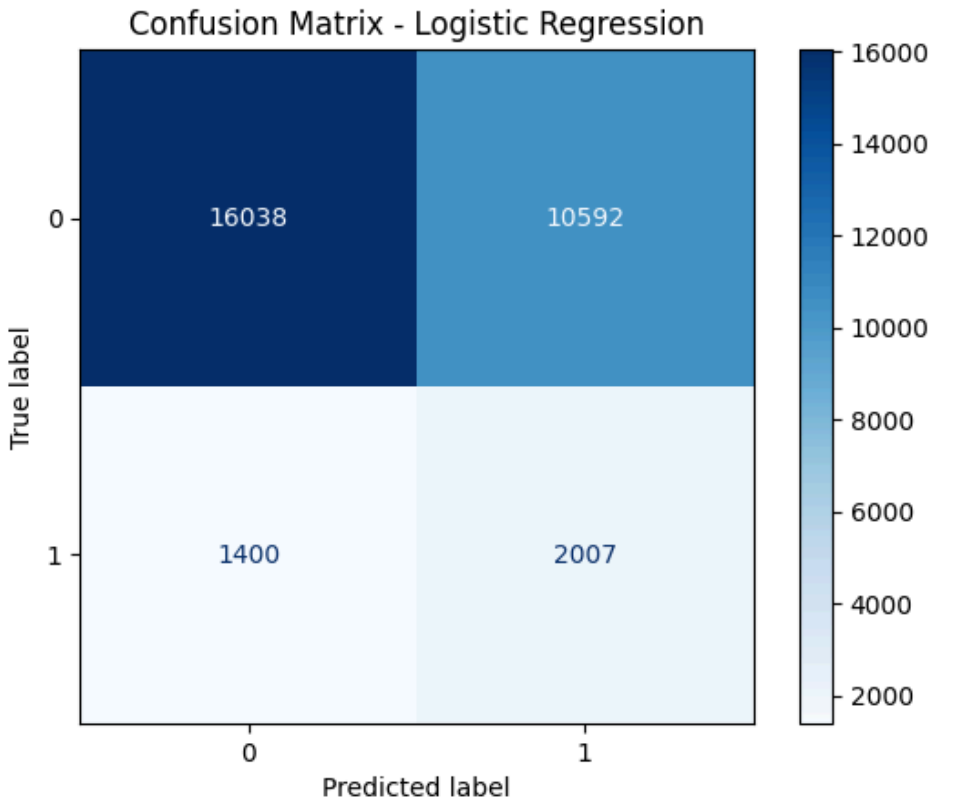
default p=0.5

	precision	recall	f1-score	support
0	0.92	0.66	0.77	26630
1	0.17	0.53	0.26	3407
accuracy			0.65	30037
macro avg	0.54	0.60	0.51	30037
weighted avg	0.83	0.65	0.71	30037

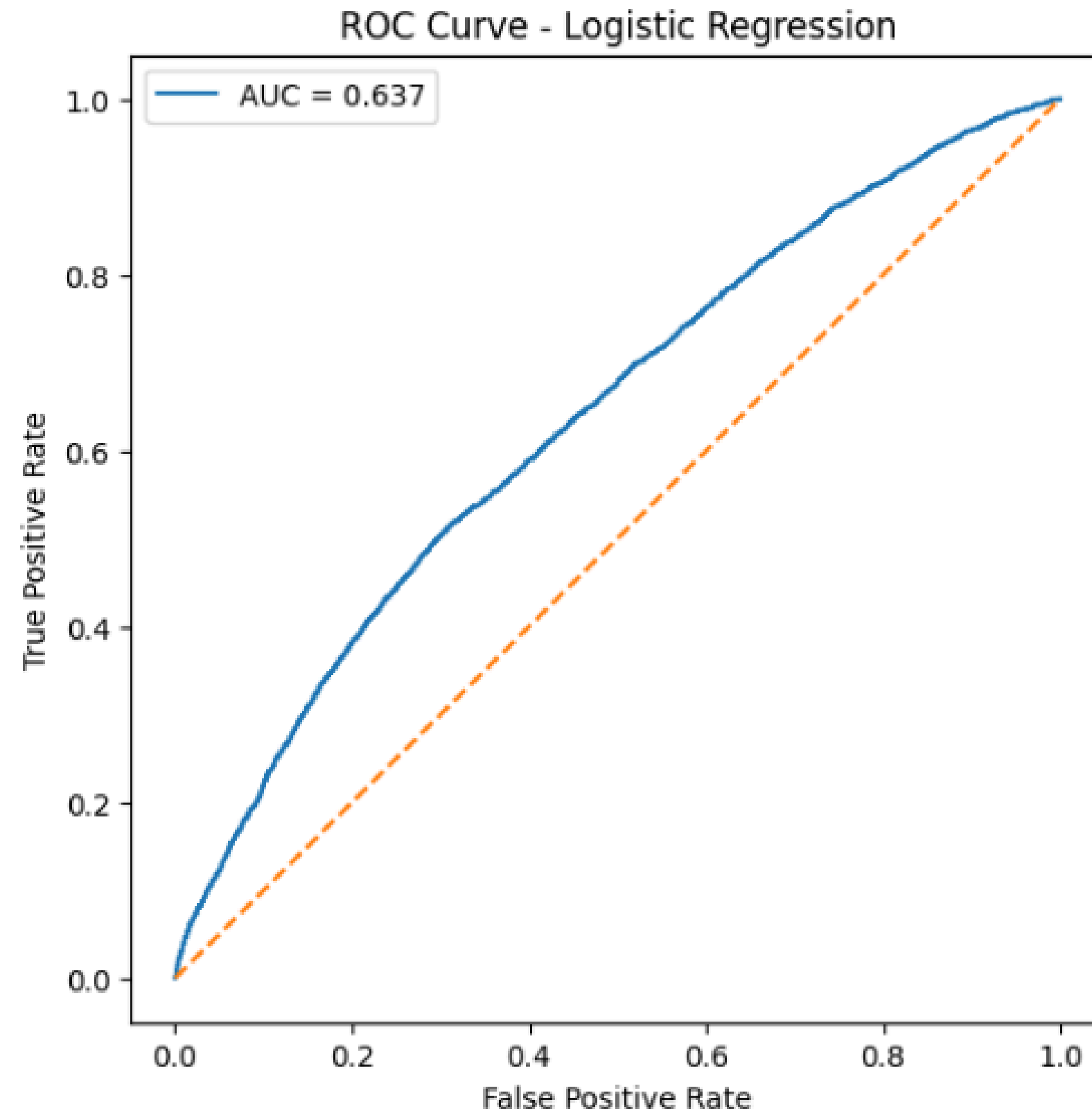


p*=0.48

	precision	recall	f1-score	support
0	0.92	0.60	0.73	26630
1	0.16	0.59	0.25	3407
accuracy			0.60	30037
macro avg	0.54	0.60	0.49	30037
weighted avg	0.83	0.60	0.67	30037



Outcomes



Reducing threshold:

- Recall for Class 1 increased: 53% → 59%
- Precision slightly decreased: 17% → 16%
- Overall accuracy decreased: 65% → 60%

==> False Negatives Decreased, False Positives increased

==> Lowering the threshold increased model sensitivity.

This aligns with the project objective:

- > Identify patients at high risk of early readmission

Random Forest:

Random Forest is a supervised machine learning algorithm used for classification and regression tasks. It is an ensemble method, meaning: It combines multiple decision trees to make a final prediction.

Early Readmission Prediction

Final Model Parameters :

- 400 trees (n_estimators=400)
- max_depth=6 (controls overfitting)
- min_samples_leaf=3 (better generalization)
- class_weight='balanced_subsample' (handles class imbalance)

Advantages of Using Random Forest :

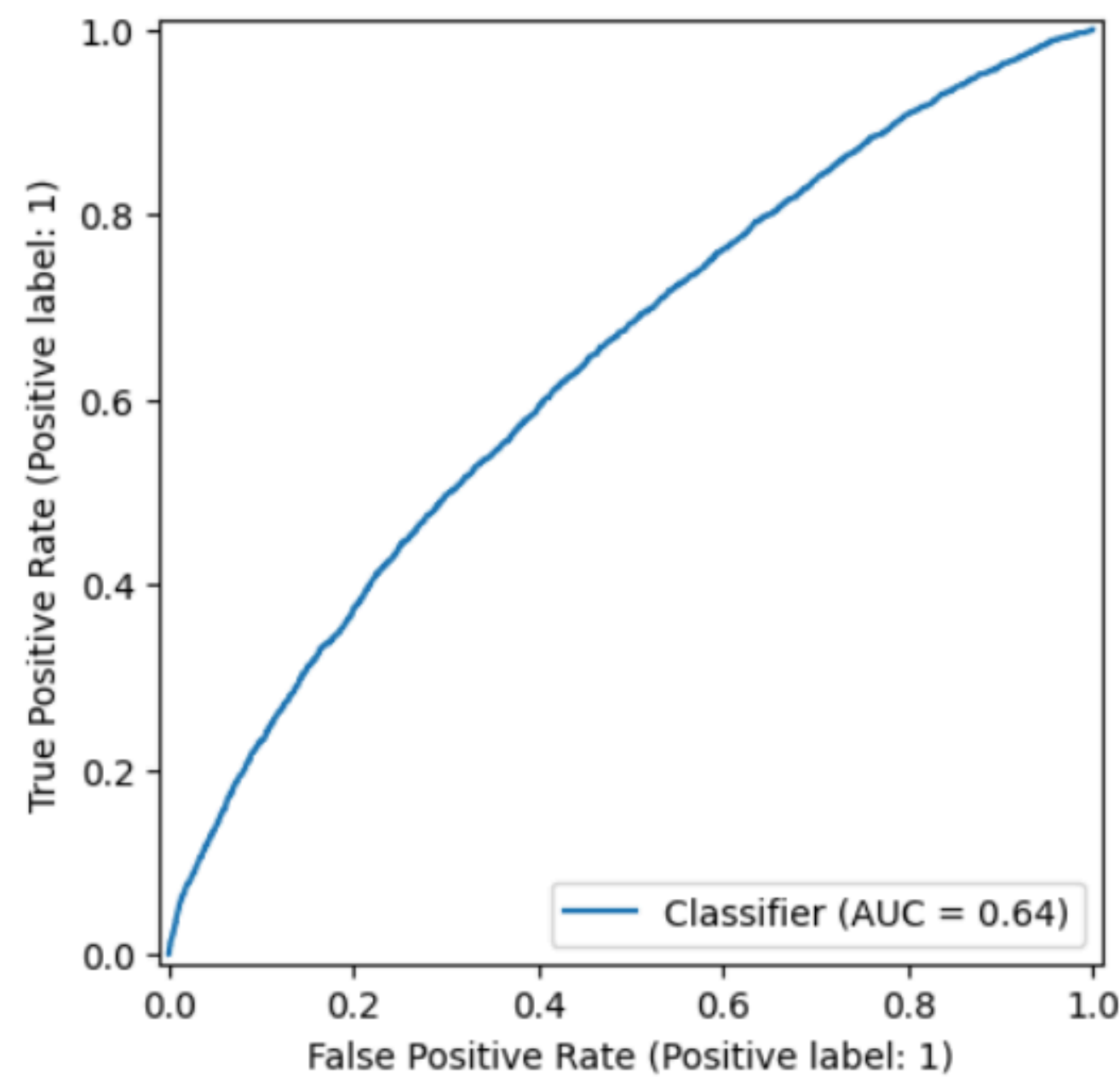
- Handles large feature sets
- Reduces overfitting
- Automatically calculates feature importance
- Works well with mixed data types

Random Forest:

Classification Report:

	precision	recall	f1-score	support
0	0.92	0.65	0.76	26630
1	0.17	0.54	0.25	3407
accuracy			0.64	30037
macro avg	0.54	0.60	0.51	30037
weighted avg	0.83	0.64	0.71	30037

ROC-AUC Score: 0.6371302770543428



Logistic coef VS Random Forest feature importance

	Feature	Coefficient	Odds_Ratio
22	medical_specialty_grouped_Psychiatry	0.423188	1.526821
24	medical_specialty_grouped_Unknown	0.364855	1.440305
19	medical_specialty_grouped_Medicine	0.333276	1.395532
23	medical_specialty_grouped_Surgery	0.305575	1.357406
3	max_glu_serum_measured	0.282175	1.326011
18	medical_specialty_grouped_Emergency	0.258861	1.295454
20	medical_specialty_grouped_Other	0.258822	1.295403
11	inpatient_ratio	0.250462	1.284618
10	total_visits	0.206606	1.229499
5	elderly	0.181122	1.198561

Random Forest Feature Importances:

	Feature	Importance
9	inpatient_ratio	0.410777
8	total_visits	0.309349
5	disease_burden	0.057447
4	elderly	0.042398
3	long_stay	0.032720
7	unstable_med	0.020351
1	diabetesMed	0.019837
6	complex_case	0.014616
13	medical_specialty_grouped_Unknown	0.012589
14	diag_1_Other	0.011570

Conclusion

- Similar predictive performance.
- Both models consistently identified inpatient_ratio, total_visits, elderly, medical_speciality as key predictors, increasing confidence in their clinical relevance
- Logistic regression tells us how each variable is affecting log-odds whereas RF only shows significant factors
- Logistic Regression preferred for interpretability and clinical inference.